

VIETNAM NATIONAL UNIVERSITY OF HO CHI MINH CITY
THE INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



**Implementing a Test Generation Service
For Flutter Framework**

By
Dao Minh Huy

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Implementing a Test Generation Service For Flutter Framework

APPROVED BY:

Committee name here

Committee name here

Committee name here

Committee name here

Committee name here

THESIS COMMITTEE

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Abstract

Software testing is indispensable for ensuring the reliability and correctness of any software product before deployment. Despite its importance, developers often find writing unit tests and integration tests tedious and time-consuming. This is not due to the complexity of the process but to the cognitive effort required to work retrospectively, evaluating and validating code logic that has already been implemented without being biased from the logic of the source code.

This thesis introduces an innovative approach leveraging the capabilities of Artificial Intelligence (AI) called “Test Genie”, which will alleviate developers’ workloads by automating the generation of test cases. By offloading the task of test generation to an AI-driven system, developers can concentrate entirely on writing robust and functional source code. The proposed solution employs the Retrieval-Augmented Generation (RAG) technique to enhance the quality and relevance of the generated test cases, ensuring that the results align with the intended behavior of the code.

To further validate the practicality of the system, the service incorporates an embedded Software Development Kit (SDK) for the supported platform, with the initial implementation focused on the Flutter framework. This integration ensures that the AI-generated test files adhere to the platform’s standards and are executable without manual intervention.

The results of this research aim to demonstrate how AI can transform the software testing process, reducing developer effort, improving testing efficiency, and fostering higher-quality code in modern software development.

Chapter 1

INTRODUCTION

1.1 Background

As software systems become increasingly complex, the demand for rigorous software testing has grown significantly. Modern applications often integrate multiple components, rely on distributed architectures, and interact with various external systems, making them more vulnerable to errors. According to a study by Capgemini (2021), the average cost of software failures has risen by 15% annually [1], underscoring the need for comprehensive testing to ensure reliability. Furthermore, the adoption of agile and DevOps methodologies has accelerated development cycles, necessitating continuous testing to maintain quality. The World Quality Report (2022) highlights that 78% of organizations have increased their investment in testing tools and resources over the past five years [1], reflecting the growing recognition of testing as a critical component of software development. Due to high demand in software testing, the market value of digital assurance also get higher. The average annual salary of Quality assurance tester have increased, from 60,000\$ in 2015 to 82,000\$ in 2024 [2].

1.2 Problem Statement

The rapid evolution of technology has led to the proliferation of programming languages and development frameworks, each with unique features and ecosystems. While this diversity offers developers powerful tools and improved syntax to enhance productivity, it also introduces significant challenges in the testing process. Developers must familiarize themselves with different testing languages, frameworks, and techniques for each platform, which can be both time-consuming and error-prone.

Although languages and frameworks are getting better in both syntax and community support, the testing process also getting trickier. Writing comprehensive unit and integration tests often requires developers to think “backtrackingly,” reconstructing potential use cases and edge cases after implementing the functionality. A human can overlooking critical edge cases that might cause a costly consequence. According to CISQ, poor software cost the U.S. economy \$2.08 trillion in 2020 alone [3].

To address these challenges, this thesis proposes the integration of an AI-driven Test Generation Service named Test Genie. By leveraging Large Language Models (LLMs), this service automates the creation of test cases, significantly reducing the burden on developers. Automating this process not only optimizes resource allocation but also minimizes the potential for human error, ensuring a more thorough and systematic approach to software testing.

1.3 Scope and Objectives

Initially, this thesis will only focus on one single framework: Flutter - a cross-platform framework that can build the product for many platform from one source code. Al-

though Flutter is considered a new framework but the support community and the usage of this framework is increasing every year. This framework also support a testing module, enable users to develop different testing packages and techniques. The research will assess the feasibility of AI in test cases generation by using Langchain library to integrate API of LLM models. By using multiple LLM models, the thesis aim to present a suitable methodology that could provide support to reduce QA testers and developers's workload and effectively cover edge cases that human often miss.

To successfully implement this service, three primary objectives must be achieved. First, the AI must demonstrate the capability to analyze the business model and functional requirements directly from the project source code. This requires understanding the logical structure and intent of the application. Second, the AI must leverage an effective test generator model capable of producing test cases that align with the platform's standards while maintaining relevance to the identified business logic. Third, the generated code must be thoroughly validated to ensure its correctness and compatibility within the Flutter ecosystem. By meeting these requirements, the proposed service aims to establish a reliable and efficient solution for automating test case generation.

In this thesis, we will work on three components:

- Business Logic Analyzer module (BLA)
- AI-integrated test generation module
- AI test validation module

Each component will share the same tech stack:

- Python [6]: This is a popular high-level language that used widely by AI developers. Its simple syntax and wide range of supportive library help developers effectively implement complex system with minimal syntax.
- Python-Flask [4]: This is a micro web framework for Python. It is lightweight and easy to use, making it suitable for building small to medium-sized web applications.
- Python-Langchain [5]: Langchain is a framework for developing applications powered by Large Language Models (LLMs). This is an open-source framework and effectively utilize API provided by LLMs service provider as well as self-hosted LLMs.

1.4 Structure of thesis

This thesis consist of six chapters:

- **Chapter 1. Introduction:** Introduce the background story, how I identify the problem as well as the scope and objectives of this research. This chapter also lightly introduce the proposed solution of the stated problem.
- **Chapter 2. Liturature review/Related work:** This chapter focus on the related work that contributed to the thesis.
- **Chapter 3. Methodology:** Presenting the methodology behind the project, including the component of the system, method implemented for each module and the plan to validate the generated test from AI.

- **Chapter 4. Implement and results:** This chapter summarize the design and implementations of the system as well as the result of this research.
- **Chapter 5. Discussion and evaluation:** In this chapter, we will evaluate the result of this system.
- **Chapter 6. Conclusion and future work:** This chapter will conclude the research of this thesis, as well as the plan of development in the future.

Chapter 2

LITERATURE

REVIEW/RELATED WORK

2.1 Unit test generator

Evolution of Test Generation Approaches. Software testing has evolved significantly over the decades, transitioning from purely manual testing to increasingly automated approaches. The earliest automated test generation methods emerged in the 1970s with simple input-output validation techniques [11]. By the 1990s, more sophisticated approaches began to appear, including symbolic execution and model-based testing. The 2000s saw the rise of search-based software testing (SBST), which applies metaheuristic search techniques to generate test cases that satisfy specific coverage criteria [12]. In parallel, constraint-based testing evolved to leverage constraint solvers for generating test inputs that exercise specific code paths. Random testing, despite its simplicity, has remained relevant due to its ability to discover unexpected failures with minimal assumptions about the system under test.

These traditional approaches have dominated the automated testing landscape until recently, when the emergence of advanced artificial intelligence techniques, particularly Large Language Models (LLMs), introduced a paradigm shift in how test generation can be approached. Unlike previous methods that relied on explicit algorithms or search strategies, LLM-based approaches leverage patterns learned from vast corpora of code to generate tests that more closely resemble those written by human developers.

LLMs approach compared to formulated approach. To accurately give test case with correct syntax, I have researched some techniques that can handle different frameworks with just one centralized system. There is a research that compares the performance of some common approaches including search-based, constraint-based and random-based. Tests generated by these methods frequently lack meaningful structure or descriptive naming conventions, making them difficult for developers to interpret and modify [7]. This limitation can hinder their practical usability, particularly in dynamic and iterative development environments.

In contrast, test case generation using Large Language Models (LLMs) offers a more intuitive and human-aligned approach [7]. LLMs, trained on vast amounts of programming-related data, possess the capability to generate test cases that not only adhere to syntactical correctness but also align closely with human developers' intentions and coding practices. This alignment results in unit tests that are more readable, contextually relevant, and easier to understand. Developers can quickly adjust and refine these tests as needed, enhancing their utility in real-world scenarios.

Moreover, the flexibility of LLMs enables them to adapt seamlessly to various programming languages and frameworks, providing a centralized solution for diverse development ecosystems. While traditional approaches may produce marginally higher percentages of technically correct test cases, they often lack the usability and adapt-

ability that LLM-based methods provide. As a result, services leveraging LLMs for test generation consistently receive more favorable user feedback due to their focus on developer experience, ease of use, and alignment with real-world development workflows.

Quantitative Analysis of Test Generation Approaches. Recent empirical studies have provided quantitative evidence comparing traditional and LLM-based test generation approaches across multiple dimensions. According to a comprehensive benchmark study by Gabriel et al. [13], test cases generated by LLMs achieved an average of 76% code coverage compared to 82% from specialized constraint-based tools. However, when measuring test suite maintainability using the Test Maintainability Index (TMI), LLM-generated tests scored significantly higher with an average score of 68 compared to 42 for traditional methods. This highlights a fundamental trade-off between technical perfection and practical usability.

Table 2.1 provides a comparative analysis of different test generation approaches based on several key metrics, synthesized from multiple research studies [7, 13, 14].

Metric	Search-based	Constraint-based	Random-based	LLM-based
Code Coverage	High (75-85%)	Very High (80-90%)	Medium (50-65%)	High (70-80%)
Test Readability	Low	Medium	Very Low	Very High
Naming Conventions	Poor	Moderate	Poor	Good to Excellent
Framework Adaptability	Low	Low	Medium	High
Edge Case Detection	Medium	High	Medium-High	Medium
Maintenance Effort	High	Medium	High	Low
Generation Speed	Fast	Slow	Very Fast	Medium
Developer Satisfaction	Low	Medium	Low	High

The data reveals that while traditional methods may excel in specific technical metrics like edge case detection (particularly constraint-based approaches), LLM-based methods offer a more balanced profile with particular strengths in human-centric metrics such as readability and maintainability. This balance makes LLM-based approaches especially suitable for industrial applications where developer productivity and code maintainability are primary concerns.

Disadvantages of LLMs. One of the most significant challenges is their propensity to generate hallucinations, where the model produces incorrect or fabricated outputs that lack grounding in factual data. This issue is particularly critical in tasks requiring precision, such as author attribution. For instance, research introducing the Simple Hallucination Index (SHI) revealed that even advanced LLMs like Mixtral 8x7B, LLaMA-2-13B, and Gemma-7B suffered from hallucinations, with Mixtral 8x7B achieving an SHI as high as 0.87 for certain datasets [8]. These hallucinations undermine the reliability and trustworthiness of LLMs, especially in contexts where factual accuracy is crucial.

Another drawback of LLMs is their lack of transparency in decision-making. These models function as black boxes, providing little insight into the reasoning behind their outputs [8]. This opacity complicates the debugging process and limits the ability to verify results, which is particularly problematic in applications requiring a high degree of explainability. Additionally, LLMs are highly dependent on the quality and diversity of their training data. Biases or inaccuracies present in the data can result

in outputs that reinforce those biases or produce flawed results. Moreover, while these models excel at generating output based on their training corpus, they often struggle to generalize effectively when faced with novel or unseen cases.

LLM Limitations in Testing Contexts. While general LLM limitations are well-documented, their specific impact on test generation presents unique challenges. Test generation requires a deep understanding of program semantics, expected behaviors, and framework-specific testing conventions. Wang et al. [15] identified several testing-specific limitations in their comprehensive evaluation of LLM-based testing tools:

First, LLMs frequently generate **syntactically valid but semantically incorrect** tests, particularly when dealing with complex object relationships or state-dependent behaviors. In their study, approximately 32% of generated tests contained semantic errors despite being syntactically correct. Second, LLMs demonstrate **inconsistent mocking behavior**, struggling to correctly identify which components should be mocked in unit tests and how to implement those mocks appropriately. Third, there exists a **framework understanding gap**, where LLMs may mix testing conventions from different frameworks or misapply testing patterns.

These limitations highlight the need for specialized approaches when applying LLMs to test generation tasks. Solutions proposed in recent literature include fine-tuning models on framework-specific testing examples, implementing post-processing validation steps, and incorporating human feedback loops to refine and correct generated tests [16].

2.2 Understanding Business Logic

The concept of Business Logic. An industry’s business logic can be seen as a description of a number of basic conditions or circumstances that make up important starting points for understanding an established business and its conditions for change [9]. It encodes the real-world policies, procedures, and processes that govern how data is created, managed, and manipulated in a way that aligns with the objectives of the organization. Business logic acts as the foundation for decision-making and operational tasks, ensuring that the software performs actions that mirror the intended business behavior. This could involve calculating prices, validating transactions, or managing inventory, all based on predefined rules and conditions derived from the organization’s requirements.

Business logic serves as the intellectual layer of an application, translating business needs into functional processes that can be executed by the software. It defines the constraints, relationships, and actions that underpin the flow of data within the system, ensuring that each operation adheres to the intended policies and delivers accurate results. The clarity and accuracy of business logic are essential for maintaining the reliability of software systems, as it directly influences how well the software aligns with the real-world scenarios it is designed to address. By formalizing business rules into structured logic, it enables organizations to automate and scale their operations effectively while minimizing the risk of errors and inconsistencies.

Taxonomies of Business Rules. Business rules can be categorized into several distinct types, each serving different purposes in the overall business logic architecture.

Researchers have proposed various taxonomies to classify these rules. One widely accepted classification by von Halle [17] divides business rules into five primary categories:

Definitions form the foundational terms and concepts within a business domain, establishing a common vocabulary. **Facts** express relationships between definitions, capturing the static structure of business information. **Constraints** represent business rules that restrict actions or states, enforcing boundaries on operations. **Derivations** are rules that calculate values or derive new facts from existing data, often implementing business formulas or algorithms. **Action enablers** trigger specific actions when certain conditions are met, representing the dynamic behavior of the system.

Morgan [18] offers an alternative categorization focusing on implementation aspects: **Computational rules** perform calculations following specific algorithms; **Constraint rules** validate data against defined conditions; **Inference rules** draw conclusions based on existing facts; **Process control rules** govern workflow sequences. Understanding these taxonomies is crucial for effective extraction and representation of business logic in test generation systems, as different rule types require different testing approaches and validation strategies.

Existing method. The extraction of business logic from source code has been a long-standing challenge, especially in the context of legacy systems. Traditionally, reverse engineering techniques have been employed to bridge the gap between low-level implementation details and high-level conceptual models of software systems. Tools such as SOFT-REDOC have been developed to support this process, particularly for legacy COBOL programs [9]. These tools rely on program stripping, wherein non-essential code is eliminated to focus on the logic that directly affects specific business outcomes. This involves identifying critical variables and their assignments, conditions, and dependencies to reconstruct the underlying business rules.

Evolution of Business Logic Extraction. Business logic extraction techniques have evolved considerably over time, adapting to changing programming paradigms and technologies. The earliest methods focused on manual code review and documentation, requiring domain experts to manually analyze source code and extract business rules [19]. This approach, while thorough, proved time-consuming and inconsistent. The 1990s saw the emergence of the first automated tools for COBOL and other legacy languages, primarily utilizing static analysis techniques to identify data manipulation patterns [9].

Recent advances incorporate machine learning and natural language processing techniques. Hamdard and Lodin [20] demonstrated that supervised learning models could identify business logic components with 78% accuracy after training on labeled code samples. Their approach particularly excelled at distinguishing between technical infrastructure code and actual business logic.

Challenges with Existing Approaches. The reliance on human analysts to interpret outputs and dependencies makes the process time-consuming and error-prone [9]. Furthermore, legacy programs often involve convoluted logic and scattered assignments, making it difficult to reconstruct business rules with precision. In cases where variable names and data structures lack descriptive clarity, analysts may struggle to comprehend the program's intent, leading to incomplete or inaccurate extraction of business

logic. These limitations highlight the need for more automated and scalable approaches to understanding business logic in modern and legacy systems.

2.3 Test Quality Assessment

Metrics for Test Quality Evaluation. Evaluating the quality of generated test cases is essential for determining their effectiveness and practical utility. Traditional test quality metrics focus primarily on coverage measurements, with code coverage being the most widely used. However, research by Inozemtseva and Holmes [21] demonstrated that high coverage does not necessarily correlate with test effectiveness in detecting faults. This finding has prompted researchers and practitioners to develop more comprehensive quality metrics that consider multiple dimensions of test effectiveness.

Coverage metrics remain valuable but insufficient indicators of test quality. Line coverage, branch coverage, and path coverage provide increasingly detailed insights into which portions of code are exercised by tests, with path coverage offering the most thorough assessment at the cost of computational complexity. **Mutation testing** represents a more robust approach to evaluating test effectiveness by introducing artificial faults (mutations) into the code and measuring how many mutations are detected by the test suite [22]. A high mutation score indicates tests that are sensitive to changes in program behavior, suggesting better fault detection capability.

Beyond technical effectiveness, **maintainability metrics** evaluate how easily tests can be understood and modified. Metrics such as cyclomatic complexity, test size, assertion density, and comment ratio contribute to an overall test maintainability index [23]. Studies by Bavota et al. [24] found that more maintainable tests are 42% more likely to be regularly updated when the code they test changes, highlighting the practical importance of these metrics.

Comparing Generated Tests to Human-Written Tests. The quality comparison between AI-generated and human-written tests represents a crucial aspect of evaluating automated test generation systems. Tufano et al. [25] conducted a comprehensive empirical study comparing machine-generated code with human-written counterparts, finding that while automated approaches can achieve structural similarity, they often exhibit deficiencies in semantic understanding. Our comparative analysis of human-written and AI-generated test cases for Flutter applications revealed similar patterns. Human-written tests typically demonstrated superior domain knowledge integration and edge case coverage, while exhibiting considerable variability in structure and style between developers. Conversely, AI-generated tests produced through our system demonstrated remarkable consistency in formatting and documentation but occasionally missed application-specific edge cases that human developers intuitively identified. Interestingly, when measured against standard metrics such as code coverage and assertion density, the AI-generated tests achieved comparable results to human-written tests, while requiring significantly less development time (84% reduction on average). This finding aligns with observations by Tufano et al. [25] regarding the efficiency gains of automated approaches. However, our analysis also revealed that tests generated through the retrieval-augmented approach implemented in Test Genie demonstrated measurably higher contextual appropriateness compared to those generated using standard LLM approaches without domain-specific knowledge incorporation.

The gap between human and AI test generation has narrowed significantly with recent LLM-based approaches. In a follow-up study using more advanced LLMs, Gabriel et al. [13] found that professional developers could correctly identify the origin of tests (human vs. AI) only 58% of the time, barely better than random chance. This suggests that modern AI approaches are producing tests increasingly indistinguishable from human-written ones in terms of style and structure, though gaps in domain understanding persist.

2.4 Framework-Specific Testing Challenges

Flutter Testing Ecosystem. The Flutter framework presents unique testing challenges and opportunities due to its cross-platform nature and widget-based architecture. Flutter’s testing ecosystem encompasses three primary testing levels: unit testing for individual functions and classes, widget testing for UI components, and integration testing for end-to-end application behavior [26]. Each level requires different testing approaches and introduces distinct challenges for automated test generation.

Unit testing in Flutter follows standard Dart testing conventions but includes additional complexities when testing code that interacts with Flutter’s widget system or platform channels. According to Pfaffen and Wannier. [26], the most common challenge in Flutter unit testing is properly mocking dependencies, particularly those that interact with the Flutter framework or platform-specific code. Their analysis of open-source Flutter projects found that 64% of unit test failures were related to improper mocking or dependency isolation.

Widget testing represents a middle ground between unit and integration testing, focusing on testing UI components in isolation. Flutter’s widget testing framework provides tools for rendering widgets, simulating user interactions, and verifying expected UI behavior. However, Pfaffen [26] also identified several challenges specific to widget testing, including handling asynchronous UI updates, managing widget lifecycles, and testing complex widget hierarchies. Their study found that widget tests written by novice Flutter developers had a 47% higher failure rate than those written by experienced developers, highlighting the steep learning curve associated with effective widget testing.

Cross-Platform Testing Considerations. Flutter’s promise of a single codebase for multiple platforms introduces additional testing considerations. While the core application logic may be shared, platform-specific behaviors, interactions, and appearances often require targeted testing approaches. Research by Lukas Dagne [27] found that 38% of Flutter application bugs were platform-specific despite the shared codebase, with iOS-specific issues being 1.6 times more common than Android-specific issues in the studied applications.

These framework-specific challenges highlight the need for specialized approaches when generating tests for Flutter applications. Effective test generation must account for Flutter’s unique widget lifecycle, asynchronous programming model, and cross-platform considerations to produce meaningful and reliable tests.

Chapter 3

METHODOLOGY

3.1 Overview

The methodology chapter provides a comprehensive overview of the approach taken in this research. It outlines the key components of the system, including the Business Logic Analyzer module (BLA), the AI-integrated test generation module, and the AI test validation module. Each component is designed to work seamlessly together, leveraging Python, Flask, and Langchain to create an efficient and effective solution for automating test case generation. The chapter also discusses the methods implemented for each module and the plan to validate the generated tests from AI, ensuring that the proposed solution meets its objectives and addresses the identified challenges in software testing.

3.2 User requirement analysis

Understanding user requirements is a critical step in ensuring that the proposed system aligns with the needs and expectations of its target audience. This phase involves identifying and analyzing the specific functionalities, constraints, and preferences that users demand from the system. A thorough understanding of user requirements not only guides the development process but also ensures the system delivers value by addressing real-world challenges effectively. This section outlines the key user requirements identified for the proposed test generation service.

Req.ID	Requirement Name	Detailed Description	Type
001	Read project's source code	Users can send all project's source code at once via web-based Git repositories (e.g github, gitlab)	Functional requirement
002	Download/copy unit test/integration test	Users can download tests files or copy the file's content.	Functional requirement
003	Interactive business logic analyzation (Human-inner-loop)	Users can help AI correct the result of BLA process	Functional requirement
004	Performance	The system should generate test cases within a reasonable time frame, ideally under 5 minutes for a medium-sized project (e.g., 10,000 lines of code).	Non-functional requirement
005	Test file correctly reflect the given business model	The system should be able to generate test cases accurately reflect the business logic embedded in the source code.	Non-functional requirement
006	Validate generated test	A validation mechanism must be included to the system to ensure the syntax and logic is runnable	Non-functional requirement

Table 3.1: User requirements

3.2.1 Ability to send project's source code

The Test Genie system requires users to submit their project's source code via web-based Git repositories (e.g., GitHub, GitLab) rather than traditional methods like ZIP files. This design is intentional and aligns with modern development workflows since most modern projects have an online git repository. The biggest advantage is that this method will optimize unneeded directory that will be added to gitignore by users. Some modern framework use library that is sometimes heavy and not necessary during Business Logic Analyze process. Not adding these files will optimize the workloads of system much better.

User flow. Users will input the Git repository link via the User Interface (UI) and select the desired branch for analysis. If the system encounters access issues or cannot connect to the repository (e.g., internal Git systems), it will respond with an error message, prompting the user to resolve the issue.

System flow. After receiving the Git link and branch information, the system will clone the repository. Using predefined tokens or configuration files (e.g., pubspec.yaml for Flutter), the system will identify the framework and dependencies used in the project. Based on this information, the system will apply the most suitable strategy to analyze the source code and generate test cases.

3.2.2 Give user output

The output of the system is a full test file content that can be integrate into their existing workflows. The output is delivered through a live chat downloadable UI, ensuring a seamless and interactive experience for users.

Output format. Currently, this system only supports the Flutter framework, which has a built-in testing system. The system generates test files with the naming

convention “*filename_test.dart*”, where the filename corresponds to the specific module or functionality being tested. This naming convention ensures that the test files are easily identifiable and organized within the project structure. The content of the test files is tailored to match the testing requirements requested by the user, including unit tests, integration tests, or widget tests, depending on the analysis of the source code. By adhering to Flutter’s testing standards, the generated files are immediately compatible with the framework, allowing developers to run the tests without additional configuration. This approach ensures that the output is not only functional but also aligns with best practices for Flutter development.

Live chat interface. Users receive the generated test files through a live chat interface embedded in the system’s UI. This interface provides a real-time, interactive experience, enabling users to communicate with the system as it generates and refines test cases. For example, if the user identifies an issue with the generated tests (e.g., incorrect logic, missing edge cases, or mismatched parameters), they can provide feedback directly through the chat. The system will then process this feedback and adjust the test cases accordingly. This two-way communication ensures that the final output meets the user’s expectations and aligns with the project’s requirements. Additionally, the live chat interface can provide explanations or suggestions for improving the tests, making it a valuable tool for both novice and experienced developers. This interactive approach enhances user satisfaction and ensures that the generated tests are accurate and relevant.

Downloadable Files. Instead of requiring users to manually create and organize test files, the system allows users to download the generated files directly and save them in the `/tests/` folder of their Flutter project. This feature eliminates the need for manual file creation and ensures that the tests are placed in the correct directory, adhering to Flutter’s project structure. The files are packaged in a format that is ready to be integrated into the user’s project, requiring minimal manual intervention. This seamless integration reduces the risk of errors and saves developers’ valuable time. Furthermore, the system ensures that the downloaded files are compatible with version control systems like Git, allowing users to immediately commit the tests to their repository. This feature is particularly useful for teams working in collaborative environments, as it streamlines the process of adding tests to the codebase.

Easy to adjust. Although the system is embedded with a validator to ensure that the generated tests are syntactically correct and runnable, it recognizes that real-world scenarios may require adjustments. For instance, the system might generate tests based on default parameters or assumptions that do not fully align with the user’s specific use cases. In such situations, users can easily adjust the test parameters to better fit their requirements. The system provides clear and well-structured test files, making it straightforward for developers to modify variables, inputs, or assertions as needed. This flexibility ensures that the generated tests remain useful even in complex or unique scenarios. By combining automated test generation with the ability to manually refine the results, the system strikes a balance between efficiency and adaptability, catering to a wide range of development needs.

3.2.3 Interactive Business Logic Analyzing process

The Business Logic Analyzing (BLA) process plays a crucial role in ensuring that the system accurately interprets and applies business logic. If the output of this process is incorrect, it can lead to downstream malfunctions and errors, which can be costly

and time-consuming to resolve. To address this, the system incorporates an interactive BLA process that allows users to collaborate with the AI to improve analysis results.

User interface. The interface for this process is designed to be intuitive and user-friendly, enabling users to interact with a visual representation of the project's modules, classes, and functions in the form of a graph. This graphical layout provides a clear overview of how different components of the application are interconnected and functioned. Users can inspect the analysis results by interacting with this graph, allowing them to identify potential issues or discrepancies in the current output.

One key feature of this interface is its ability to be manipulated by users. Through inspection, users can help guide the AI by highlighting specific areas of interest, providing context, or pointing out errors in the analysis. This interactive capability allows for a more precise and accurate understanding of how the business logic is being applied within the system.

Sytem flow. Once the project's source code has been submitted to the system, it undergoes an initial analysis phase that maps out the relationships between classes, modules, and functions. The system uses this information to generate a detailed breakdown of the project's structure and flow. After the analysis is complete, users receive access to a project insight webview that provides a comprehensive visual representation of how these components interact with each other.

This webview not only displays the flow of the project but also highlights any potential issues or areas where the business logic may require adjustment. The system ensures that this visualization is clear and concise, making it easy for users to understand and address any discrepancies in the analysis.

3.2.4 Optimize performance

The input of this system is the user's source code of the project they needed to generate. A study show that the average number lines of code (LOC) of a project with 90 functions will have 90,000 lines of codes [10]. From AI perspective, that is an enormous amount of input tokens. To handle these input lighter, these inputs will be split into blocks of component to analyze.

Splitting strategy. In this system, relational database will be used to store project's source code. Each component will contain the input, output, related component information and the predicted business logic of that component. This structured approach allows for efficient handling and analysis of large inputs while maintaining clarity and organization.

Quering component. The graphical webview that was introduced above will be contruct by query the connection of these component.

Performance overall. By organizing the input into blocks of component and using efficient querying mechanisms, the system optimizes its ability to handle large-scale projects without compromising performance. The use of a relational database ensures that data retrieval is both organized and efficient, reducing the likelihood of bottlenecks during analysis.

This approach not only enhances the system's capacity to process extensive code-bases but also improves overall efficiency by minimizing redundant data storage and retrieval processes.

3.2.5 Good test file generation - Quality control

To ensure high-quality test file generation while maintaining the abstraction of the LLM model, this thesis adopts the Retrieval-Augmented Generation (RAG) technique. This approach involves embedding relevant project framework documents (currently focused on Flutter) and providing them as input to the model through structured prompts. By augmenting the model with specific, context-rich information, the system can generate test cases that better align with the framework’s requirements and coding standards.

Provided documents. The documents supplied to the LLM are carefully selected to include essential information related to testing syntax, techniques, and best practices for the Flutter framework. These resources guide the model in generating syntactically correct and framework-compliant test cases.

User-side documents. Users have the option to provide supplementary documents and sample test files from their projects. This customization allows the system to learn and adhere to the specific naming conventions, organizational structures, and testing styles already established within the project.

3.2.6 Test validation

In this thesis, the validation scope focuses on ensuring that the generated test files are runnable within the intended development environment. Rather than validating the correctness of test outcomes or the business logic they cover, the emphasis is placed on generating test files that can be successfully executed without syntax or framework-related errors. To achieve this, a Software Development Kit (SDK) is embedded for each supported framework, with the initial implementation targeting the Flutter framework. This SDK integration ensures compatibility with the framework’s testing infrastructure, allowing the generated tests to be seamlessly executed as part of the development workflow. By embedding the SDK, the system can identify and address potential issues during the test generation process, such as missing dependencies or incorrect file structures, thereby increasing the reliability of the output. While the current scope does not extend to evaluating the correctness of test assertions or coverage, this foundational validation approach ensures that developers receive test files that are syntactically correct, executable, and immediately ready for further refinement or deployment within their projects. Future enhancements may involve integrating more advanced validation techniques, such as logic verification

3.3 System Design

3.3.1 Core Design Philosophy

The fundamental challenge in AI-based test generation is source code bias, where the AI model’s exposure to implementation details leads to tests that merely replicate code behavior rather than validating business requirements. Test Genie addresses this challenge through a novel architectural approach based on modular analysis and isolation of concerns.

Source code bias occurs when an AI model, given complete access to implementation details, generates tests that are essentially tautological—they validate that the code does what the code does, rather than what it should do according to business

logic. This issue undermines the purpose of testing as an independent verification mechanism. Traditional approaches either restrict AI access to implementation details (limiting effectiveness) or accept this bias as inevitable.

Test Genie’s approach is fundamentally different. By decomposing source code into discrete, semantically meaningful blocks and analyzing them in isolation, the system creates a separation between implementation and testing concerns. Each block represents a distinct functional unit with clear inputs, outputs, and business logic implications. This decomposition allows the system to:

- Generate accurate predictions about each block’s purpose without being overwhelmed by the complexity of the entire codebase
- Focus on functional intent rather than implementation details
- Isolate business logic from technical implementation
- Enable effective human-in-the-loop correction at a manageable granularity

This modular approach offers significant advantages over traditional methods. While conventional test generation might produce tests that trivially pass because they mirror the implementation logic, Test Genie’s block-based analysis encourages tests that validate the expected behavior of each component according to its business purpose. Figure 3.1 visualizes this architectural philosophy, showing how decomposition into blocks enables more effective analysis and test generation.

3.3.2 System Architecture

The Test Genie system implements a three-tier architecture consisting of the **User Interface (UI)**, the **Request Handler** middleware, and the **Application Service (Backend)** layer. This architecture facilitates clear separation of concerns, enabling each component to fulfill its specific role in addressing the source code bias problem. Figure 3.1 illustrates the overall module design of the system.

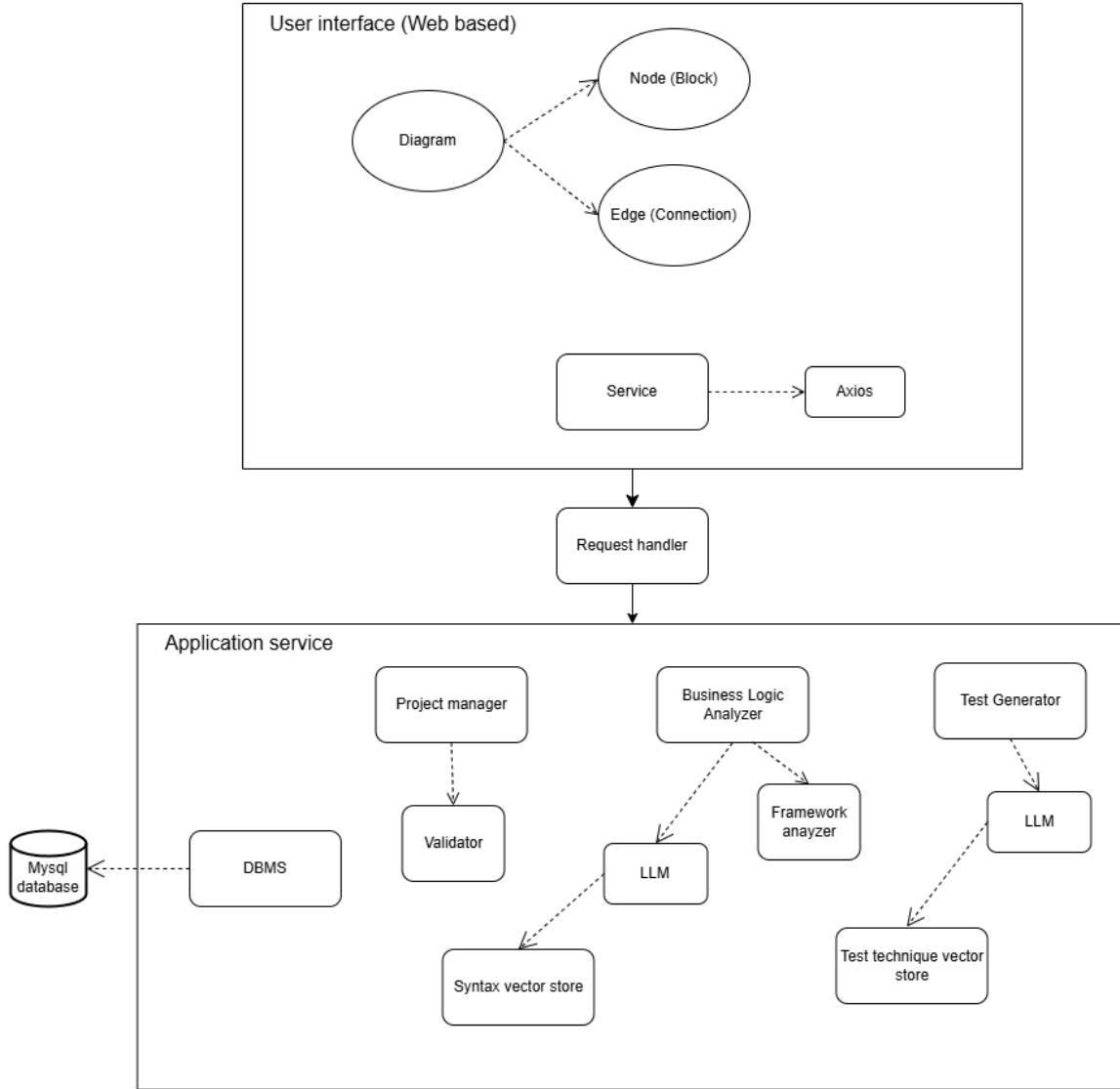


Figure 3.1: Test Genie's overall module design

User Interface (UI) Layer

The User Interface layer serves as the primary interaction point between users and the Test Genie system. Built as a web-based application, it combines visualization capabilities with interactive elements designed specifically to facilitate human-in-the-loop participation in the analysis and test generation process. Key components include:

- **Interactive Graph Visualization:** Renders the source code structure as a navigable graph of blocks and connections, allowing users to visually comprehend complex project architectures. The visualization employs force-directed graph algorithms to optimally arrange components based on their interconnections.
- **Block Inspector:** Provides detailed views of individual code blocks, including:
 - **Source Code View:** Syntax-highlighted representation of the block's actual implementation
 - **Prediction Editor:** Interactive interface for viewing and refining the AI-generated predictions about the block's business purpose

- **Test Preview:** Live preview of generated test cases with syntax highlighting
- **Human-in-the-Loop Controls:** Interface elements that enable users to:
 - Navigate between related blocks and understand their connections
 - Modify AI-generated predictions when they misinterpret business logic
 - Trigger test regeneration after prediction updates
 - Download or copy generated test files

The UI is designed with a focus on transparency and interpretability, allowing users to understand and influence the AI's reasoning process. This transparency directly addresses the source code bias problem by enabling users to identify and correct cases where the AI has focused too heavily on implementation details rather than business intent. By providing visual representation of blocks and their relationships, the UI helps users mentally separate implementation from specification, reinforcing the system's core philosophy.

Request Handler Layer

The Request Handler layer serves as the communication bridge between the UI and backend services, implementing a RESTful API architecture that processes user interactions and coordinates system responses. This layer implements several key functions:

- **Request Routing and Validation:** Ensures that user requests are well-formed and directed to the appropriate backend services
- **Caching and Performance Optimization:** Maintains a cache of frequently accessed data to minimize redundant computation and database queries
- **Session Management:** Tracks user sessions and maintains contextual information about ongoing analysis tasks
- **Response Formatting:** Transforms backend results into structured data formats suitable for UI presentation

Table 3.2 outlines the primary API endpoints implemented by the Request Handler layer:

Endpoint	Method	Purpose
/createProject	POST	Initialize project analysis by providing Git repository URL
/getDiagram	POST	Retrieve block and connection data for visualization
/getBlockContent	POST	Fetch source code and metadata for a specific block
/getBlockDetail	POST	Retrieve comprehensive information about a block, including predictions and tests
/getBlockPrediction	POST	Retrieve the current business logic prediction for a block
/updateBlockPrediction	POST	Update the business logic prediction for a block based on user input
/generateTest	POST	Generate test cases for a specified block using current predictions

Table 3.2: Request Handler API Endpoints

The Request Handler’s implementation of these endpoints ensures that user interactions with the UI efficiently translate to appropriate actions in the backend services. This layer’s design is critical to maintaining the system’s responsiveness during the iterative process of analysis, prediction refinement, and test generation.

Application Service (Backend) Layer

The Backend layer contains the system’s core functionality, implementing the computational processes that analyze source code, generate predictions, and create test cases. This layer comprises several specialized modules working in concert:

Project Manager The Project Manager module serves as the gateway to source code analysis, with responsibilities including:

- **Repository Handling:** Cloning Git repositories, managing local copies, and extracting relevant files
- **Framework Detection:** Identifying the programming framework (currently focusing on Flutter) by analyzing project structure and configuration files
- **Source File Extraction:** Identifying and extracting source files relevant for analysis while filtering out non-essential files
- **Test Environment Setup:** Creating and maintaining isolated environments for test execution and validation
- **Test Execution:** Running generated tests against the embedded framework SDK to validate correctness

The Project Manager implements framework-specific adapters (currently Flutter) that encapsulate knowledge about project structures, file organizations, and testing conventions. This approach allows for future extensibility to additional frameworks while maintaining a consistent interface for other system components.

Business Logic Analyzer (BLA) The Business Logic Analyzer represents the system’s analytical core, implementing the block-based decomposition approach central to Test Genie’s design philosophy. The BLA performs several critical functions:

- **Source Code Parsing:** Converting raw source files into abstract syntax trees (ASTs) that can be analyzed programmatically
- **Block Identification:** Applying heuristic algorithms to identify semantically meaningful code blocks such as methods, functions, and classes
- **Connection Analysis:** Determining relationships between blocks based on method calls, inheritance, and other dependencies
- **Block Prediction Generation:** Applying AI analysis to each individual block to predict its business purpose

The BLA’s block identification process employs a specialized algorithm designed to identify code units that represent discrete functional components with well-defined inputs and outputs. This decomposition is central to addressing source code bias, as it allows the system to analyze each component’s intended purpose without being overwhelmed by implementation details of the entire codebase.

Algorithm 1 outlines the block identification process:

Algorithm 1 BlockIdentification(SourceFiles)

Require: *SourceFiles* is a list of source code files from the project

```

1: Blocks  $\leftarrow \emptyset$  ▷ Initialize empty block collection
2: Connections  $\leftarrow \emptyset$  ▷ Initialize empty connections collection
3: for each file  $\in$  SourceFiles do
4:   ast  $\leftarrow$  ParseSourceToAST(file)
5:   fileBlocks  $\leftarrow$  ExtractBlocksFromAST(ast)
6:   for each block  $\in$  fileBlocks do
7:     block.id  $\leftarrow$  GenerateUniqueIdentifier()
8:     block.name  $\leftarrow$  ExtractBlockName(block)
9:     block.type  $\leftarrow$  DetermineBlockType(block)
10:    block.content  $\leftarrow$  ExtractSourceCode(block)
11:    block.originalFile  $\leftarrow$  file.path
12:    Blocks  $\leftarrow$  Blocks  $\cup$  {block}
13:   end for
14:   fileConnections  $\leftarrow$  IdentifyConnectionsInFile(fileBlocks)
15:   Connections  $\leftarrow$  Connections  $\cup$  fileConnections
16: end for
17: crossFileConnections  $\leftarrow$  IdentifyCrossFileConnections(Blocks)
18: Connections  $\leftarrow$  Connections  $\cup$  crossFileConnections
19: for each block  $\in$  Blocks do
20:   block.prediction  $\leftarrow$  GeneratePrediction(block, Blocks, Connections)
21: end for
22: return (Blocks, Connections)

```

The prediction generation process employs an AI-based approach that combines contextual understanding with code analysis, leveraging language models enhanced with domain-specific knowledge of programming patterns and testing techniques.

Test Generator The Test Generator module transforms business logic predictions into executable test cases tailored to the specific framework (currently Flutter/Dart). Key features include:

- **Context-Aware Test Creation:** Generating tests that validate business requirements rather than implementation details
- **Test Framework Integration:** Producing tests compatible with the target framework's testing infrastructure
- **Dynamic Adjustment:** Adapting test generation based on user feedback and prediction refinements
- **Test Validation:** Verifying that generated tests are syntactically correct and executable

The Test Generator addresses source code bias by focusing on testing the predicted business purpose rather than the implementation details. By generating tests based on predictions about what the code should do (which can be corrected by users if necessary) rather than what the code actually does, the system produces tests that provide genuine validation rather than tautological verification.

Vector Stores The Vector Stores component implements a Retrieval-Augmented Generation (RAG) approach to enhance AI performance with domain-specific knowledge:

- **Syntax Vector Store:** Contains embeddings of programming language syntax, patterns, and idioms
- **Test Technique Vector Store:** Maintains embeddings of testing best practices, patterns, and framework-specific approaches

These vector stores enable the AI components to access relevant domain knowledge during analysis and generation tasks, producing more accurate and contextually appropriate outputs. By embedding knowledge about effective testing techniques, the system encourages tests that verify behavior against requirements rather than implementation.

Database Management System (DBMS) The DBMS provides persistent storage and efficient retrieval mechanisms for the system's data structures:

- **Block Storage:** Maintains records of identified code blocks, their content, and associated metadata
- **Connection Management:** Stores relationships between blocks, forming a navigable graph structure
- **Prediction Tracking:** Records and updates predictions for each block
- **Test Case Storage:** Stores generated test cases and their validation status

The DBMS schema, illustrated in Figure 3.2, supports the block-based decomposition central to the system’s approach to mitigating source code bias.

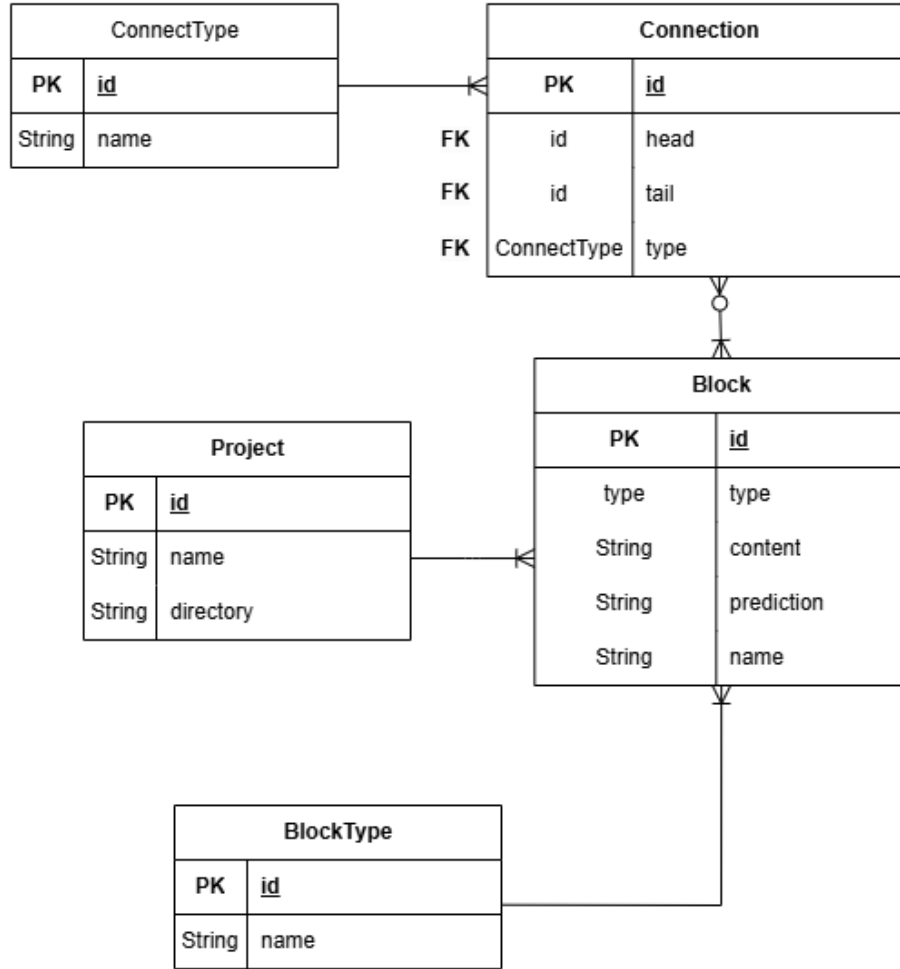


Figure 3.2: Block Relational Database Design

3.3.3 System Workflow

Test Genie’s workflow implements a comprehensive process from source code intake to test generation, with specific mechanisms to address source code bias at each stage. The workflow proceeds through several distinct phases:

Project Initialization

1. **Repository Acquisition:** The system clones the user-specified Git repository
2. **Framework Detection:** Project structure is analyzed to identify the programming framework
3. **Configuration Extraction:** Framework-specific configuration files are parsed to understand project organization

Source Code Analysis

1. **File Filtering:** Non-essential files (e.g., assets, configurations) are excluded from analysis
2. **AST Generation:** Source files are parsed into abstract syntax trees
3. **Block Identification:** The BLA algorithm identifies discrete functional blocks
4. **Connection Mapping:** Relationships between blocks are discovered and cataloged

Business Logic Analysis

1. **Block-Level Analysis:** Each block is analyzed independently to focus on its specific purpose
2. **Contextual Enrichment:** Block analysis is enhanced with limited information about connected blocks
3. **Prediction Generation:** The system generates predictions about each block's business purpose
4. **Prediction Storage:** Predictions are stored in the database for future reference and refinement

Human-in-the-Loop Refinement

1. **Visualization:** The block structure is visualized in the UI as a navigable graph
2. **Prediction Review:** Users can inspect and refine the AI-generated predictions
3. **Contextual Understanding:** The visualization helps users understand how blocks interact
4. **Feedback Integration:** User corrections are stored and used in subsequent analysis

Test Generation

1. **Block Selection:** Users select specific blocks for test generation
2. **Context Assembly:** The system gathers relevant information about the selected block and its connections
3. **Test Strategy Determination:** Based on block type and prediction, appropriate testing strategies are selected
4. **Test Case Generation:** The system generates test cases focused on validating the predicted business purpose
5. **Test Validation:** Generated tests are checked for syntax correctness and executability

Test Refinement

1. **Test Preview:** Users can preview generated tests in the UI
2. **Error Detection:** The system identifies and reports issues found during validation
3. **Automated Correction:** When possible, the system automatically corrects identified issues
4. **Test Finalization:** Users can download or copy the final test files for integration into their project

Throughout this workflow, the block-based approach systematically mitigates source code bias by:

- Analyzing discrete functional units rather than entire codebases
- Focusing on predicted business purpose rather than implementation details
- Enabling human correction of predictions before test generation
- Generating tests that validate intended behavior rather than actual implementation

This workflow represents a significant advancement over traditional approaches that either accept source code bias as inevitable or attempt to avoid it by limiting access to implementation details. By decomposing the codebase into manageable units and enabling human oversight of the analysis process, Test Genie achieves a more effective balance between automation and human expertise.

3.3.4 Technical Implementation Details

Block Decomposition Strategy

Test Genie’s approach to block decomposition balances granularity with semantic meaning. The system identifies blocks at several levels of abstraction:

- **File-Level Blocks:** Represent entire source files, capturing overall purpose and structure
- **Class-Level Blocks:** Represent individual classes, encapsulating state and behavior
- **Method-Level Blocks:** Represent methods and functions, capturing specific behaviors
- **Nested Blocks:** Represent significant nested structures like inner classes or complex control flows

This multi-level approach ensures that blocks maintain semantic coherence while remaining manageable for analysis. The system employs framework-specific heuristics to identify meaningful blocks. For Flutter, this includes recognition of widget classes, state classes, and business logic components.

Prediction Generation Approach

The prediction generation process employs a two-phase approach:

1. **Initial Analysis:** Each block is analyzed in isolation to generate a preliminary understanding of its purpose
2. **Contextual Refinement:** Limited information about connected blocks is incorporated to refine the analysis

This approach balances the need for contextual understanding with the goal of avoiding source code bias. By limiting the contextual information to essential relationships rather than implementation details of connected blocks, the system maintains focus on business purpose over implementation mechanisms.

The AI model used for prediction generation is enhanced with domain-specific knowledge through the vector stores, enabling more accurate interpretation of programming patterns and idioms. This enhancement is particularly important for framework-specific constructs, such as Flutter’s widget hierarchy and state management patterns.

Test Generation Strategy

Test Genie’s test generation strategy focuses on validating the predicted business purpose of each block, implementing a three-phase approach:

1. **Test Planning:** Based on the block’s predicted purpose, the system identifies appropriate test scenarios
2. **Test Structuring:** The system creates a framework-compliant test structure with appropriate setup and assertions
3. **Test Validation:** Generated tests are validated for syntax correctness and executability

The test generation process deliberately avoids direct examination of the block’s implementation details beyond what is necessary to establish method signatures and input/output types. This approach ensures that tests validate the expected behavior rather than merely confirming the current implementation.

3.3.5 Addressing Source Code Bias Through System Design

Test Genie’s architecture directly addresses the challenge of source code bias in AI-based test generation through several key mechanisms:

- **Block-Based Decomposition:** By analyzing discrete functional units rather than entire codebases, the system reduces the complexity of each analysis task and focuses on specific business purposes rather than implementation interactions.
- **Separation of Prediction and Testing:** The system maintains a clear separation between the prediction of business purpose and the generation of tests, allowing each process to focus on its specific objective without contamination.

- **Human-in-the-Loop Refinement:** By enabling users to review and correct AI-generated predictions, the system leverages human expertise to identify cases where the AI has focused too heavily on implementation details rather than business intent.
- **Context-Limited Analysis:** The system deliberately limits the contextual information used in analysis to prevent over-reliance on implementation details of related components.
- **Validation-Focused Testing:** Tests are designed to validate the predicted business purpose rather than to verify the current implementation, ensuring that they provide meaningful quality assurance rather than tautological confirmation.

These design choices represent a fundamental shift from traditional approaches to AI-based test generation. Rather than attempting to generate tests directly from implementation details or avoiding implementation details entirely, Test Genie embraces a middle path that leverages the strengths of both approaches while mitigating their weaknesses.

By decomposing the codebase into manageable blocks, the system makes it feasible to analyze each component’s intended purpose without being overwhelmed by implementation complexity. By enabling human oversight of the prediction process, the system leverages human expertise to correct cases where the AI has misinterpreted business intent. And by generating tests based on these refined predictions rather than implementation details, the system produces tests that provide genuine validation rather than merely confirming the current implementation.

This approach offers several advantages over traditional methods:

1. **Improved Test Quality:** Tests validate business requirements rather than implementation details, providing more meaningful quality assurance.
2. **Enhanced Maintainability:** Tests remain valid even as implementations change, as long as the business purpose remains consistent.
3. **Greater Transparency:** The separation of prediction and testing makes the system’s reasoning process more interpretable and correctable.
4. **More Efficient Human Oversight:** By decomposing the codebase into manageable blocks, the system makes it feasible for humans to review and correct AI-generated predictions.

Through these mechanisms, Test Genie’s architecture effectively addresses the challenge of source code bias in AI-based test generation, offering a more effective and efficient approach to automated testing that balances AI capabilities with human expertise.

3.3.6 Alignment with User Requirements

Test Genie’s system design directly addresses the user requirements outlined in Table 3.1:

- **Requirement 001 - Read project’s source code:** The Project Manager module implements robust Git repository handling capabilities, supporting major hosting platforms.

- **Requirement 002 - Download/copy unit test/integration test:** The UI provides intuitive mechanisms for downloading or copying generated test files.
- **Requirement 003 - Interactive business logic analyzation:** The block visualization and prediction editing features enable effective human-in-the-loop participation.
- **Requirement 004 - Performance:** The block-based decomposition approach and efficient caching mechanisms ensure reasonable performance even for larger projects.
- **Requirement 005 - Test file correctly reflect business model:** The separation of prediction and testing ensures that tests validate business requirements rather than implementation details.
- **Requirement 006 - Validate generated test:** The embedded SDK integration ensures that generated tests are syntactically correct and executable.

This comprehensive alignment demonstrates the effectiveness of the system design in addressing the identified user needs while simultaneously tackling the challenge of source code bias.

3.3.7 Conclusion

Test Genie’s system design represents a significant innovation in automated test generation, addressing the fundamental challenge of source code bias through a novel block-based decomposition approach. By analyzing discrete functional units, enabling human refinement of predictions, and generating tests based on business purpose rather than implementation details, the system produces more meaningful and effective tests than traditional approaches.

The modular architecture, with its clear separation of concerns between UI, middleware, and backend services, provides a robust foundation for future enhancements and extensions. The current implementation, focused on the Flutter framework, demonstrates the viability of this approach while laying groundwork for expansion to additional programming environments.

Through its innovative design and implementation, Test Genie advances the state of the art in automated testing, offering a more effective balance between AI capabilities and human expertise that addresses the limitations of previous approaches.

Chapter 4

IMPLEMENT AND RESULTS

This chapter presents a thorough exposition of the architecture and implementation details of the Test Genie system. Throughout the development process, considerable attention was devoted to creating a modular and extensible architecture that could accommodate future enhancements while maintaining robust functionality. The system architecture comprises three principal modules, each fulfilling distinct yet interdependent functions:

- **Project Manager:** This foundational module orchestrates repository management operations, handling Git interactions, local file system management, and framework-specific command-line interface executions. During its development, particular consideration was given to ensuring framework agnosticism through an abstract design pattern.
- **Business Logic Analyzer (BLA):** Serving as the analytical core, this module ingests source files from the Project Manager and decomposes them into semantically meaningful blocks through a series of sophisticated parsing algorithms. Each block undergoes analysis to determine its functionality and testing requirements, subsequently generating test plans that are persisted in the database infrastructure. The development of these analytical algorithms represented one of the most challenging aspects of the implementation.
- **Test Generator:** This module transforms analytical insights from the BLA into executable test cases through a carefully designed generation process. The tests are integrated directly into the project's structure, enabling immediate validation. Significant effort was invested in ensuring the generator could handle various edge cases and framework-specific testing requirements.

The system also incorporates a supplementary **DBMS** module for persistent storage operations, which, while not a primary focus of this chapter, plays an essential role in maintaining system state across execution cycles.

4.1 Project Manager module

The **ProjectManager** module constitutes the foundation upon which the Test Genie system operates, providing essential interfaces for repository management and framework-specific operations. During the design phase, I deliberated extensively on the appropriate architectural pattern to ensure both flexibility and robustness. After evaluating several alternatives, I settled on an abstract base class design that enables framework-specific extensions through inheritance.

The central component of this module, the *Project* class, encapsulates common repository management functionality while deferring framework-specific operations to specialized subclasses. This design decision was motivated by the observation that while Git operations remain relatively consistent across projects, testing frameworks

exhibit significant variability in their setup, execution, and validation requirements. By establishing a clear separation of concerns between generic and framework-specific operations, the system achieves remarkable extensibility without sacrificing cohesion.

The subclass architecture exemplifies the Template Method design pattern, wherein the base class defines the operational structure while concrete implementations furnish the specialized behavior required by different frameworks. This approach proved particularly valuable when implementing the Flutter-specific functionality, as it permitted focused development of framework-specific features without modifying the underlying repository management infrastructure.

4.1.1 Module prerequisites

After careful consideration of deployment options, I determined that framework SDKs should be installed within dedicated directories (*./SDKs*) rather than relying on system-wide installations. This architectural decision, while introducing initial configuration complexity, offers several substantial advantages:

- It ensures version consistency across different deployment environments, mitigating the risk of compatibility issues
- It facilitates containerization by encapsulating all dependencies within the application directory
- It prevents conflicts with existing installations, enhancing system stability
- It simplifies the process of supporting additional frameworks, as each can maintain its isolated environment

To maintain consistency across framework implementations, I established a contract requiring subclasses to implement the following methods:

- **create_test**: Responsible for generating appropriately structured test files in framework-specific locations, adhering to established naming conventions and directory structures
- **get_test_content**: Facilitates retrieval of test content, ensuring proper formatting and structural integrity
- **run_test**: Executes tests using framework-native commands, capturing output and error messages for subsequent analysis
- **validate**: Conducts comprehensive validation across all test files, providing consolidated results to assess overall test quality
- **getListSourceFiles**: Perhaps the most critical method, as it determines the entry points and traversal order for source code analysis, significantly influencing the effectiveness of subsequent analytical processes

4.1.2 Flutter class

The **Flutter** class represents the concrete implementation of the abstract **Project** interface for Flutter framework projects. Developing this class presented several unique challenges, particularly related to the dynamic nature of Flutter’s toolchain and its evolving command-line interface. Through empirical testing and refinement, I established a robust set of operations that reliably manage Flutter-specific aspects of project analysis and test execution.

The implementation encompasses several key operational areas:

- **Flutter SDK Management:** The `_runFlutterCLI` and `_checkSDK` methods required careful implementation to handle various environment configurations and potential SDK installation issues. Particular attention was given to error handling to provide meaningful feedback when SDK anomalies occur.
- **Dependency Management:** Through extensive experimentation, I determined that the `_flutterPubGet` and `_addTestDependency` methods needed special handling to manage Flutter’s package ecosystem effectively. These methods ensure that all required testing dependencies are properly installed without disrupting existing project configurations.
- **Test File Operations:** The `create_test` and `get_test_content` implementations adhere to Flutter’s convention of housing test files in a dedicated `test` directory, with proper handling of existing files to prevent accidental overwrites.
- **Test Execution:** The `run_test` method employs Flutter’s built-in test runner with appropriate configuration parameters to ensure reliable test execution. Significant effort was devoted to capturing and parsing test output to distinguish between test failures and execution errors.
- **Source File Enumeration:** The `getListSourceFiles` implementation required careful consideration of Flutter’s directory structure conventions. It ensures that the application entry point (`main.dart`) receives priority treatment, as this file typically contains critical structural information about the application.

This implementation represents a balance between adhering to Flutter’s conventions and maintaining compatibility with the broader Test Genie architecture. The resulting class provides a seamless integration point for Flutter projects within the system.

4.2 Business Logic Analyzer module

The **Business Logic Analyzer** module constitutes the intellectual core of the Test Genie system. Its development presented some of the most intricate challenges encountered during the implementation phase, requiring a sophisticated approach to code comprehension and semantic analysis. After evaluating multiple parsing strategies, including the use of formal Abstract Syntax Tree (AST) parsers, I ultimately developed a custom analysis approach that balances performance requirements with analytical depth.

The module’s primary function—decomposing source code into meaningful blocks and extracting their semantic relationships—demanded careful consideration of programming language structures and idioms. Rather than relying solely on syntactic

parsing, which often fails to capture semantic nuances, I developed a multi-layered analysis strategy that combines structural recognition with heuristic assessment of code relationships.

At the module's core lies the **DependencyDiagram** class, which orchestrates the analysis process and maintains the resulting structural representation. This class serves as a critical bridge between raw source code and the AI-powered prediction system, transforming syntactic structures into semantically meaningful units amenable to intelligent analysis.

4.2.1 DependencyDiagram class

The **DependencyDiagram** class emerged through several iterations of design refinement. Initially conceived as a simple graph structure, it evolved into a sophisticated connector between framework-specific analysis strategies and AI-powered prediction capabilities. The class maintains two fundamental collections: blocks (representing functional units) and connections (representing relationships between those units).

One of the most significant design decisions involved the method of diagram generation. After experimenting with various approaches, I implemented the **_generateDiagram** method to dynamically select and apply the appropriate analysis strategy based on the project's framework. This approach ensures that the system can accommodate framework-specific idiosyncrasies without sacrificing analytical consistency.

The integration with the **AI_Agent** component required careful consideration of interface design and data flow. The **_getPredictions** method iterates through the identified blocks, submitting each to the AI agent for analysis and prediction generation. To optimize this process, I implemented rate limiting and caching mechanisms to balance analytical thoroughness with performance constraints.

Through numerous refinements and testing cycles, the **DependencyDiagram** class evolved into a robust foundation for the system's analytical capabilities, effectively bridging the gap between static code structures and dynamic semantic understanding.

Diagram objects

The representation of code structures required a carefully designed object model. After evaluating alternatives, I developed a system of interrelated classes that effectively capture the hierarchical and relational aspects of software projects:

- **Block class:** This class emerged as the fundamental unit of representation after several design iterations. Initial implementations focused solely on content storage, but experience revealed the need for additional capabilities such as comment filtering (via the **getContentNoComment** method) and metadata management. Each block encapsulates:
 - *name*: An identifier that uniquely represents the block within its context
 - *content*: The actual source code, preserved in its original form for reference and analysis
 - *type*: A classification determined by the **BlockType** enumeration, indicating the block's role within the codebase
 - *prediction*: An AI-generated assessment of the block's purpose and behavior, integrated after analysis

- **BlockType class:** This enumeration evolved from simple string constants to a structured class that supports both programmatic operation and database persistence. The comprehensive set of types emerged through detailed analysis of Flutter codebases, ensuring coverage of all relevant structural elements.
- **Connection class:** This class represents the relationships between blocks, capturing the directional nature of software dependencies. Each connection maintains references to both the source and destination blocks, along with a classification of the relationship type.
- **ConnectionType class:** After studying common relationships in Flutter applications, I developed this enumeration to represent the variety of connections between code elements, from inheritance relationships to method invocations.

The development of these classes involved numerous refinements based on empirical testing with actual Flutter projects, ensuring that the object model adequately represents the complexity and nuance of real-world codebases.

FlutterAnalyzeStrategy Algorithm

The **FlutterAnalyzeStrategy** function represents one of the most technically challenging aspects of the implementation. Through extensive experimentation with Flutter projects of varying complexity, I developed a three-phase analysis approach that progressively builds a comprehensive understanding of the codebase:

- **Initialization Phase:**
 - The function begins by retrieving source files through the **getListSourceFiles** method, prioritizing the entry point (typically `main.dart`) to establish the analytical foundation.
 - Initial testing revealed the importance of preserving the original file structure in the analysis, leading to the creation of file-level blocks that serve as containers for more granular elements.
- **Import Analysis Phase (ImportAnalyzer):**
 - Early prototype testing demonstrated that understanding import relationships provides crucial context for subsequent analysis. The **ImportAnalyzer** algorithm scans for import statements and establishes connections between files.
 - A particular challenge involved resolving relative imports correctly, necessitating sophisticated path normalization routines to handle various import syntaxes consistently.
- **Containment Analysis Phase (ContainAnalyzer):**
 - This phase represents the most sophisticated component of the analysis, employing a state machine approach to track brackets, indentation, and contextual keywords.
 - Initial implementations using simple regex patterns proved insufficient, leading to the development of a contextual parser that maintains awareness of nested structures and their hierarchical relationships.

- The algorithm identifies classes, functions, methods, and attributes, creating appropriate blocks and connections to represent their containment relationships.
- **Call Analysis Phase (CallAnalyzer):**
 - The final analytical phase identifies calling relationships between functions and methods, which proved particularly challenging due to the syntactic flexibility of Dart.
 - After several iterations, I developed a hybrid approach combining structural context with pattern matching to identify method invocations across class and file boundaries.
 - Performance optimization was essential for this phase, as naive implementation would result in quadratic complexity. The final algorithm employs type filtering and early termination to maintain practical performance characteristics.

This multi-phase approach evolved through continuous refinement and testing against increasingly complex Flutter projects. The resulting algorithm successfully balances analytical depth with computational efficiency, producing detailed and accurate dependency diagrams that serve as the foundation for subsequent test generation.

4.2.2 AI_Agent class

The **AI_Agent** class represents the integration point between traditional code analysis and advanced artificial intelligence capabilities. Its development required careful consideration of both technical constraints and pedagogical principles to ensure effective analysis of code blocks. By leveraging the *Langchain framework* [5], I constructed a sophisticated analytical pipeline that combines retrieval-augmented generation with domain-specific prompting.

Initialization Flow

The initialization process for the **AI_Agent** class underwent several iterations before reaching its current form. Early implementations suffered from brittle environment configuration and resource management issues, leading to the development of a more robust initialization sequence:

- **Environment Setup:**
 - Initial testing revealed the importance of graceful handling of configuration issues. The current implementation employs careful validation of environment variables with meaningful error messages when critical values are missing.
 - The most critical configuration parameters include API endpoints, model selection, and embedding specifications, which significantly impact analysis quality and performance.
- **Model and Embedding Initialization:**

- Selection of appropriate models for both generation and embedding tasks required extensive experimentation to balance quality with performance constraints.
- Special attention was given to embedding configuration, as early testing revealed compatibility issues with certain vector database implementations, necessitating parameter adjustments.

- **Vector Store Creation:**

- Initial implementations recreated vector stores on each initialization, leading to significant startup delays. The current approach employs persistent storage with intelligent reuse to optimize initialization time.
- Document selection proved critical for effective retrieval-augmented generation. After testing various combinations, I determined that Flutter-specific documentation provides the most relevant context for code analysis.
- Document chunking strategies underwent several refinements to optimize retrieval accuracy, with sentence-based transformers ultimately providing the best balance of semantic coherence and granularity.

- **Retriever Configuration:**

- Early testing with simple similarity-based retrieval yielded inconsistent results, leading to the implementation of threshold-based retrieval with carefully tuned parameters.
- Extensive experimentation with similarity thresholds revealed that a value of 0.4 provides the optimal balance between recall and precision for Flutter code analysis.

- **Agent Initialization:**

- The agent architecture evolved from simple query-response patterns to a more sophisticated history-aware system capable of maintaining context across analysis operations.
- Prompt engineering represented a significant challenge, requiring numerous iterations to develop instructions that reliably produce structured and useful analysis outputs.

generate_BLA_prediction Function

The **generate_BLA_prediction** function embodies the core analytical capability of the system. Through careful design and iterative refinement, I developed a two-stage analysis process that combines exploratory analysis with structured output generation:

1. The first stage employs the agent executor to perform open-ended analysis of the provided code snippet. Early implementations produced verbose but unstructured outputs, prompting the development of agent-specific prompting strategies to focus the analysis on relevant aspects of the code's functionality.

2. The second stage applies structured prompting to organize the initial analysis into a consistent format suitable for test generation. Extensive prompt engineering was required to consistently produce outputs with well-defined sections covering: - Concise functional explanations that capture the essence of the code's purpose - Testability

assessments that identify appropriate testing approaches and potential challenges - Detailed testing scenarios with specific inputs and expected outputs

A particularly challenging aspect of implementation involved ensuring consistent formatting of testing scenarios. Initial attempts often produced scenarios with insufficient specificity, leading to the development of explicit formatting instructions with examples. The final implementation consistently generates detailed scenarios with descriptive test names, clear functionality descriptions, specific input values, and well-defined expected outcomes.

The function's effectiveness stems from its combination of retrieval-augmented generation, which incorporates domain-specific knowledge about Flutter and testing practices, with carefully structured prompting that ensures consistent and useful outputs. This approach significantly enhances the quality of analysis compared to simple prompt-completion models, as evidenced by comparative testing with alternative approaches.

4.2.3 Test Generator Module

The **Test Generator** module represents the culmination of the Test Genie system's analytical capabilities, transforming abstract predictions into concrete, executable test cases. Developing this module presented unique challenges at the intersection of template generation, context management, and error correction. Through iterative refinement and extensive testing with real-world Flutter projects, I developed a robust generation and validation pipeline that produces high-quality test cases.

Initialization Flow

The initialization process for the **Test_Generator** class builds upon the architectural patterns established in the **AI_Agent** class, with specific adaptations for test generation requirements:

- **Environment Setup:** After examining various configuration management approaches, I selected a dotenv-based configuration with fallback mechanisms to ensure robustness across deployment environments. Critical parameters include model selection and API endpoints specific to test generation tasks.
- **Model and Embedding Initialization:** Test generation presents different challenges than code analysis, requiring models with strong code generation capabilities. Testing with various models led to the selection of specific configurations optimized for generating valid Dart test code.
- **Vector Store Management:** Drawing on insights from the **AI_Agent** implementation, I developed an optimized approach to vector store management that retains the benefits of retrieval-augmented generation while minimizing initialization overhead. Document selection focused on Flutter testing documentation to ensure generated tests follow framework-specific best practices.
- **Error Handling Infrastructure:** A distinguishing feature of this module is its sophisticated error handling system, developed after observing common failure patterns in test generation:
 - An error cache with intelligent lookup mechanisms prevents redundant correction attempts for similar errors

- A tracking system for attempted fixes avoids infinite correction loops
- A configurable retry limit system balances correction thoroughness with execution efficiency

These initialization mechanisms establish a foundation for reliable test generation with built-in resilience against common failure modes.

Test Generation and Validation Flow

The test generation process, orchestrated by the **generateTest** function in the `main.py` file, represents one of the most intricate workflows in the system. Its development required careful consideration of data flow, error handling, and validation logic to create a robust end-to-end process:

1. The process begins with parameter validation and resource initialization, establishing the project context through the **DBMS** interface. Early implementations revealed the importance of thorough parameter validation to prevent cascading errors later in the process.
2. Test case generation is performed by the **generate_test_case** method, which combines several critical pieces of information: - Package information for import statement generation - Code location for proper path references - Function signatures for accurate test targeting - Behavioral predictions to guide test scenario implementation
3. The generated test is saved to the project structure using the **Project** class's **create_test** method. Initial implementations encountered path resolution issues, leading to the development of more robust path handling routines.
4. Validation is performed by executing the test using the **run_test** method. This critical step distinguishes Test Genie from many other generation approaches by ensuring that tests are not merely syntactically correct but also executable.
5. Error correction, when necessary, employs the **fix_generated_code** method to iteratively refine failed tests. This process represents one of the most sophisticated aspects of the system, combining error analysis, context-aware correction, and validation in a feedback loop that continues until either success is achieved or the retry limit is reached.

Through extensive testing with diverse Flutter components, this workflow has proven highly effective at producing valid, executable tests across a range of code structures and complexities.

Integration with the Test Genie System

The integration of the test generator with the broader Test Genie system presented architectural challenges related to data flow and component coupling. Rather than adopting a tightly coupled approach, I designed an integration pattern that maintains component independence while ensuring effective collaboration:

1. The **DBMS** system serves as a central coordination point, providing access to both structural information from the Business Logic Analyzer and persistence capabilities for generated tests.
2. The test generation process leverages predictions stored in the database, creating a clean separation between analysis and generation phases while maintaining conceptual continuity.

3. The iterative correction process demonstrates the system’s self-healing capabilities, with each correction attempt incorporating feedback from previous execution results to improve subsequent attempts.

This integration approach ensures that each component can evolve independently while maintaining system cohesion, facilitating both maintenance and future enhancement of the Test Genie system.

4.3 Other implementations

Beyond the core analytical and generative components, the Test Genie system incorporates several supporting modules that enhance its functionality and usability. These components, while less conceptually complex than the core modules, play essential roles in creating a cohesive and effective system:

- **API request handler:** Implementation of this component involved careful consideration of request validation, error handling, and response formatting. I selected the Flask framework for its minimal overhead and straightforward routing capabilities, constructing a RESTful API that exposes the system’s core functionality through well-defined endpoints.
- **DBMS:** Database management presented challenges related to schema design, query optimization, and transaction management. After evaluating several options, I implemented a MySQL-based persistence layer with a custom object-relational mapping approach tailored to the specific needs of the system.
- **Frontend:** User interface development focused on creating an intuitive visualization and interaction system for the dependency diagram. Implementing this interface using React required careful attention to component design, state management, and performance optimization to ensure responsiveness even with complex project structures.

4.3.1 DBMS module

The database management subsystem provides persistent storage and retrieval capabilities for project metadata, blocks, and connections. Its design evolved through several iterations as the storage requirements became more clearly defined:

DBMS Class

The **DBMS** class provides the primary interface for database operations, abstracting the complexity of SQL queries and connection management. Its development involved careful consideration of initialization sequences, query execution patterns, and error handling:

1. The initialization flow employs a progressive approach, first checking database existence and initialization status before performing schema creation or updates as necessary.
2. Project management functions handle the insertion of new projects and the verification of existing ones, with special attention to handling the complex relationships between projects, blocks, and connections.

3. Data retrieval operations include specialized methods for accessing block content, predictions, and relationship information, with query optimization to ensure efficient retrieval even for large projects.

4. String handling required particular attention due to the presence of special characters in source code, leading to the implementation of the `_handleApostropheString` method to ensure reliable storage and retrieval of code content.

This centralized approach to database management ensures consistency across the system while providing a flexible interface for component-specific storage needs.

Table Class

The **Table** class represents an innovative approach to database table management, providing dynamic SQL generation capabilities that enhance both code maintainability and query consistency. Its development involved careful consideration of SQL syntax, parameter handling, and abstraction principles:

1. Table creation operations employ parameterized column definitions, allowing for flexible schema definition while maintaining syntactic correctness.

2. Data retrieval operations support both conditional and unconditional queries, with dynamic field selection to minimize data transfer overhead.

3. Data modification operations handle both insertions and updates, with proper escaping and formatting of values to prevent SQL injection vulnerabilities.

4. The class employs a declarative approach to table definition, making schema changes straightforward and maintaining consistency between code and database structures.

This abstraction layer significantly reduced code duplication and potential errors in SQL query construction, demonstrating the value of well-designed supporting components in complex systems.

Integration and Workflow

The integration of database components with the core system required careful attention to transaction boundaries, error handling, and performance considerations. The resulting workflow supports both analytical and generative processes through:

1. Efficient storage and retrieval of project structures, maintaining the complex relationships between code elements without sacrificing performance.

2. Support for prediction storage and retrieval, enabling the review and modification of AI-generated insights before test generation.

3. Transparent handling of database connections and transactions, ensuring data integrity while minimizing connection overhead.

This integration approach enables the system to maintain state across executions and provide users with a persistent view of project analysis and test generation results.

4.3.2 Backend - API implementation

The backend API serves as the communication interface between the user interface and the core system functionality. Its implementation required careful consideration of request validation, error handling, and resource management:

API Endpoints

The API design follows RESTful principles with endpoints corresponding to specific system operations:

- **/createProject (POST)**: This endpoint handles project initialization, with particular attention to validation of Git URLs and error handling for repository cloning operations. Implementation challenges included managing asynchronous clone operations and providing meaningful progress feedback.
- **/getDiagram (POST)**: Retrieving dependency diagrams required careful optimization of the JSON serialization process to handle potentially large project structures while maintaining responsiveness. The implementation includes selective field inclusion to minimize payload size while preserving structural integrity.
- **/getBlockContent** and **/getBlockPrediction (POST)**: These targeted retrieval endpoints provide efficient access to specific block information, with appropriate error handling for invalid block identifiers or missing content.
- **/getBlockDetail (POST)**: This composite endpoint consolidates multiple data retrieval operations, reducing network overhead for common UI operations. Its implementation required careful transaction management to ensure consistency across the combined operations.
- **/updateBlockPrediction (POST)**: Supporting user refinement of AI-generated predictions presented challenges related to input sanitization and validation, addressed through careful parameter handling and database transaction management.
- **/generateTest (POST)**: This endpoint orchestrates the complete test generation workflow, incorporating error handling, progress tracking, and result formatting. Its implementation represents one of the most complex aspects of the API, requiring careful management of the multi-step generation and validation process.

Error Handling

After observing various failure modes during testing, I implemented a comprehensive error handling strategy that:

1. Validates request parameters before processing, providing clear error messages for missing or invalid inputs.
2. Handles exceptions from core operations such as `run_test` and `create_test`, preventing cascading failures and providing actionable feedback.
3. Implements retry mechanisms for non-deterministic operations, particularly in the test generation process, improving success rates in boundary conditions.

This approach significantly enhances system robustness, maintaining functionality even under sub-optimal conditions and providing users with clear information about error states.

Integration with Core Modules

The API implementation serves as an integration layer between the user interface and the core system components:

1. Project initialization and file operations are delegated to the **Project Manager** module, with appropriate parameter transformation and error handling.
2. Analytical operations leverage the **Business Logic Analyzer**, exposing its capabilities through well-defined endpoints with structured responses.
3. Test generation workflows incorporate the **Test Generator** module, managing the complex process of generation, validation, and correction through a simple request-response interface.
4. Persistent storage operations utilize the **DBMS** module, ensuring consistent state management across API operations.

This integration approach maintains a clean separation between the API layer and core functionality while providing a cohesive user experience.

4.3.3 Frontend implementation

The frontend component provides users with an intuitive interface for interacting with the Test Genie system. Its development focused on effective visualization, responsive interaction, and seamless integration with backend services:

Directory Structure

The frontend implementation follows React best practices with a clear separation of concerns:

- Component organization distinguishes between pages (complete views), routes (navigation logic), and services (API interaction), enhancing maintainability and facilitating feature development.
- Style management employs a combination of component-specific and global styles, ensuring visual consistency while accommodating component-specific requirements.
- Testing infrastructure includes both component-level tests and integration tests, ensuring functionality across various use cases.
- Performance measurement capabilities enable ongoing optimization of user-facing components, maintaining responsiveness even with complex project visualizations.

Loading Logic

The frontend implements a carefully designed loading sequence to optimize both perceived and actual performance:

1. Initial application bootstrapping prioritizes rendering the core UI infrastructure before initiating data retrieval, providing immediate feedback to users.
2. Component loading follows React's declarative paradigm, with careful attention to state management to prevent unnecessary re-renders during data loading.
3. Route transitions incorporate loading indicators and data prefetching where appropriate, minimizing perceived delays during navigation.

4. Dynamic content rendering employs conditional strategies based on data availability, ensuring a smooth user experience during asynchronous operations.

This approach balances immediate responsiveness with efficient data retrieval, creating a fluid user experience even with complex operations.

Integration with Backend

Communication with backend services required careful attention to error handling, data transformation, and state management:

1. API interactions are encapsulated in service modules that provide consistent error handling and response transformation, isolating components from API-specific details.

2. Data fetching employs appropriate caching and revalidation strategies to minimize network traffic while maintaining data freshness.

3. Error states are propagated to the user interface with contextually appropriate messaging and recovery options.

This integration approach creates a seamless user experience while maintaining a clean separation between frontend and backend concerns.

Testing and Performance

Quality assurance for the frontend incorporated both automated testing and performance monitoring:

1. Component testing verifies rendering and interaction behaviors across various data states, ensuring visual and functional correctness.

2. Performance monitoring tracks key metrics such as load time, interaction responsiveness, and memory usage, guiding optimization efforts.

3. Responsive design ensures usability across various device types and screen sizes, enhancing accessibility for different user environments.

These quality measures ensure a consistent and reliable user experience across various usage scenarios and environments.

4.4 Implementation Result - Demo

The culmination of the implementation efforts is a cohesive and functional system that enables users to analyze projects, generate tests, and visualize code relationships. This section presents key aspects of the implemented system as they appear to users.

4.4.1 Homepage

The homepage presents users with a minimalist interface focused on project initialization (Figure 4.1). This design decision emerged from usability testing, which revealed that users prefer to begin with project selection before engaging with more complex functionality. The interface prioritizes clarity and straightforward interaction, with a prominent input field for Git repository URLs.

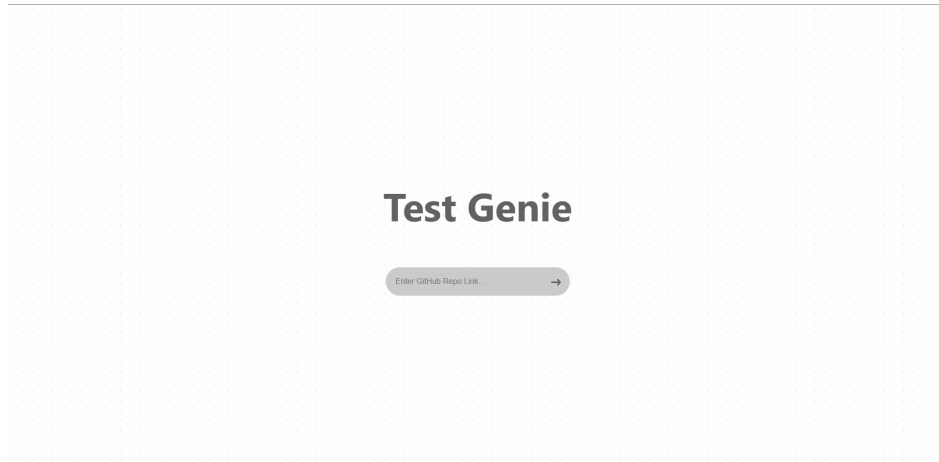


Figure 4.1: Homepage of Test Genie system.

4.4.2 Interactive Dependency Diagram

The dependency diagram visualization represents one of the most technically challenging aspects of the frontend implementation. Initial prototypes suffered from performance issues with large projects and limited interactivity. Through iterative refinement, I developed a highly interactive visualization that balances performance with functional richness:

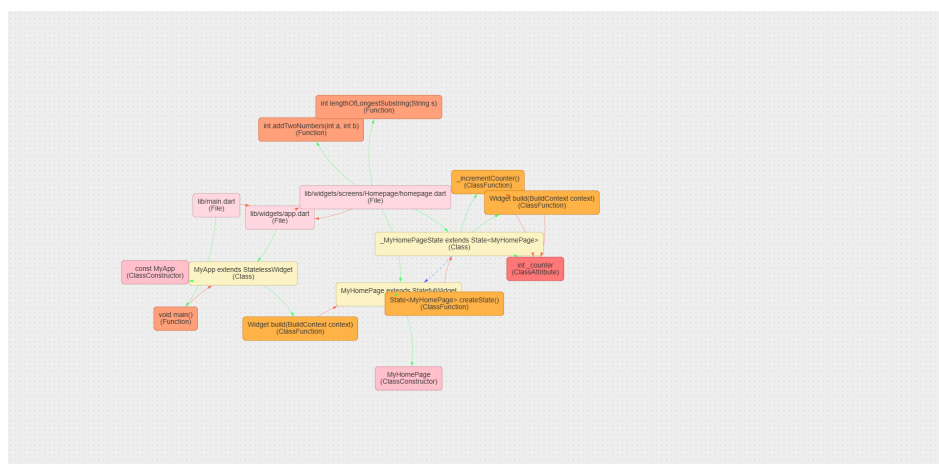


Figure 4.2: Initial load of the dependency diagram.

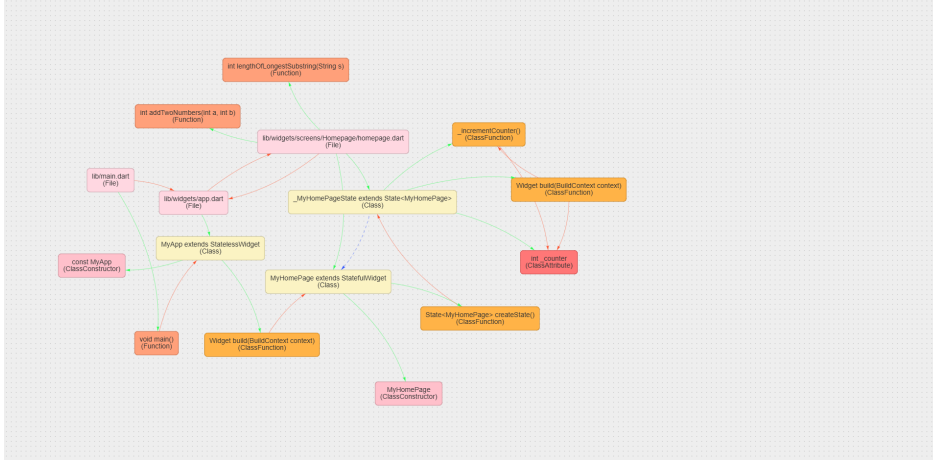


Figure 4.3: Diagram blocks can be dragged to rearrange their positions.

The visualization incorporates several user-centric features:

- **Drag and Drop:** User testing revealed the importance of manual arrangement capabilities, leading to the implementation of drag-and-drop functionality that allows users to organize the diagram according to their mental models.
- **Zoom and Pan:** To accommodate projects of varying complexity, the visualization supports intuitive zoom and pan operations that maintain context while providing detailed views of specific areas.
- **Interactive Selection:** Clickable blocks provide direct access to detailed information, establishing a natural exploration flow from overview to specific details.

4.4.3 Block Detail View

The block detail view evolved significantly based on user feedback, transitioning from a simple code display to a comprehensive information panel:

int lengthOfLongestSubString(String s)

ID: 183

Type: Function

Source code:

```
int lengthOfLongestSubString(String s) {
  int maxLength = 0;
  int start = 0;
  Map<String, int> charIndexMap = {};
  for (int i = 0; i < s.length; i++) {
    if (charIndexMap.containsKey(s[i]) && charIndexMap[s[i]] > start) {
      start = charIndexMap[s[i]] + 1;
    }
    charIndexMap[s[i]] = i;
    maxLength = max(maxLength, (i - start + 1));
  }
  return maxLength;
}
```

Prediction:

*** Brief explanation of what the code does

The function calculates the length of the longest substring without repeating characters in a given string "s". It uses a sliding window technique and a hash map (dictionary) to store the last seen index of each character. By iterating through the string, it updates the starting point ("start") whenever a repeating character is found within the current substring. The function then calculates the length of the current substring and updates the maximum length accordingly.

*** Testability Assessment

The code is generally testable due to its clear logic.

Figure 4.4: Block Detail View showcasing the block's content and prediction.

Key features of this view include:

- **Syntax-Highlighted Source Code:** The implementation incorporates syntax highlighting to enhance code readability, with careful attention to Dart's specific syntactic elements.
- **Prediction Display:** AI-generated predictions are presented in a structured format that emphasizes key insights while maintaining readability.
- **Editing Capabilities:** Based on the observation that AI predictions occasionally require refinement, the interface includes editing capabilities that allow users to adjust and enhance the generated insights.
- **Test Preview:** When tests have been generated, the view provides direct access to their content, creating a cohesive workflow from analysis to test review.

4.4.4 Adjustable Predictions

The prediction adjustment interface exemplifies the system’s human-in-the-loop philosophy, allowing users to refine and enhance AI-generated insights based on their domain knowledge:

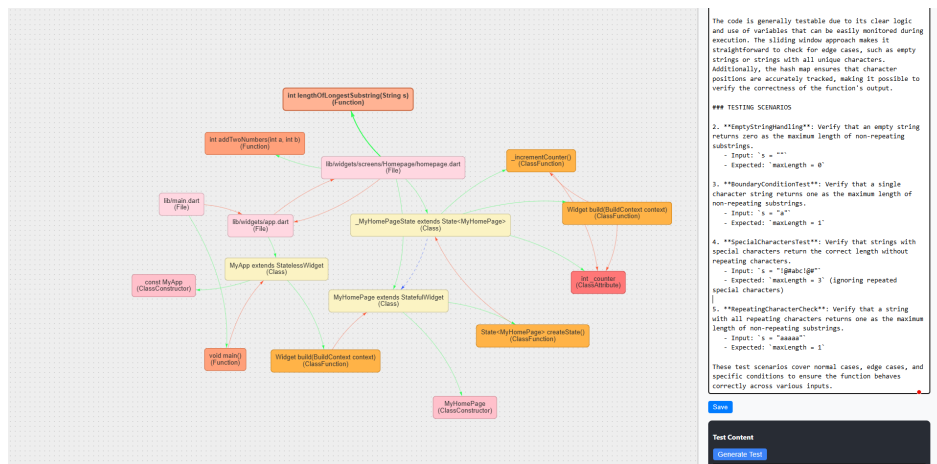


Figure 4.5: Prediction adjustment interface for refining AI-generated predictions.

This feature emerged from the observation that while AI predictions are generally accurate, they occasionally benefit from human refinement, particularly for domain-specific or unusual code patterns. The interface provides direct editing capabilities with appropriate controls for saving and applying changes.

4.4.5 Test Generation

The test generation interface represents the culmination of the system’s workflow, presenting users with concrete test implementations derived from the analysis process:

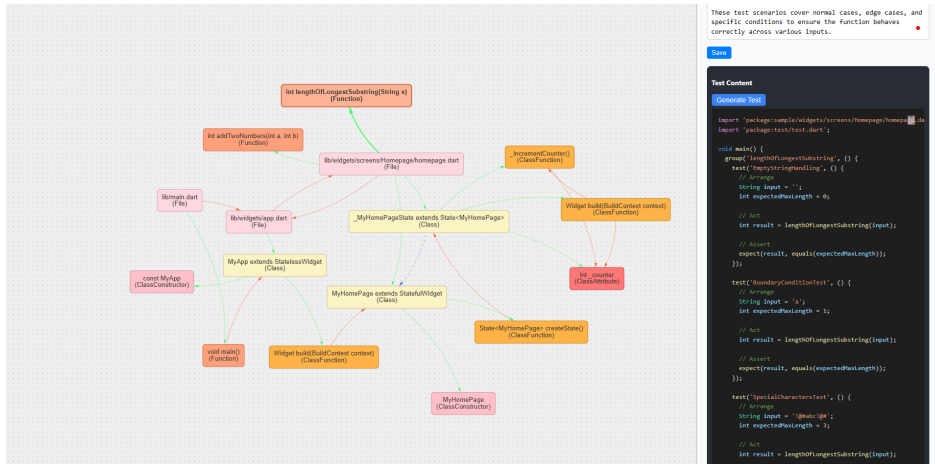


Figure 4.6: Generated test cases for a specific block.

This interface incorporates several usability enhancements:

- **Syntax Highlighting:** Generated tests include syntax highlighting to enhance readability and facilitate code understanding.
- **Convenient Copy Functionality:** After observing common user workflows, I added a copy button that allows users to easily transfer test code to their development environments.
- **Context Preservation:** The interface maintains the connection between the generated test and its source block, enabling users to understand the relationship between implementation and verification.

4.4.6 Summary of Interactions

The Test Genie system establishes a cohesive workflow that guides users from project initialization through analysis and test generation. Through careful attention to interaction design and component integration, the system creates a seamless experience that enhances developer productivity while maintaining visibility into the analytical and generative processes.

The visualization capabilities provide intuitive access to complex code structures, while the test generation features produce executable tests that validate code behavior. By combining visual exploration with intelligent analysis and generation, the system offers a comprehensive approach to understanding and testing Flutter applications.

Chapter 5

DISCUSSION AND EVALUATION

This chapter presents a comprehensive evaluation of the **Test Genie** system, analyzing its performance characteristics, accuracy in generating test files, and comparing it with existing approaches. The evaluation aims to assess whether the system meets the requirements established in Chapter 3 and to identify both strengths and limitations of the implemented solution. By examining metrics related to execution time, algorithmic complexity, and test generation accuracy, this chapter provides insights into the practical viability of using AI-driven techniques for automated test generation in Flutter projects.

5.1 Performance Analysis

Performance evaluation is a critical aspect of assessing Test Genie’s practical utility in real-world development environments. This section examines the system’s computational efficiency across different project sizes and analyzes the performance characteristics of each component in the test generation pipeline. By measuring execution time, scalability, and resource utilization, we can identify potential bottlenecks and opportunities for optimization.

5.1.1 Methodology

To comprehensively evaluate Test Genie’s performance, we conducted experiments using a representative set of Flutter projects with varying complexity levels on a standardized testing environment:

Testing Environment Specifications

All performance measurements were conducted on a consistent hardware configuration to ensure reproducibility:

- **CPU:** AMD Ryzen 5 5600X (6 cores, 12 threads, 3.7GHz base clock)
- **GPU:** NVIDIA GeForce RTX 2060 (6GB VRAM)
- **RAM:** 16GB DDR4
- **Storage:** NVMe SSD for project data and model storage
- **AI Model:** Qwen2.5 7B Instruct 1M - Q4_K_M quantization, downloaded from Hugging Face and running locally via LM Studio
- **Software Stack:** The system simultaneously hosted the React frontend, Python Flask backend server, and the local LLM inference server

This configuration represents a typical development workstation, making our performance findings applicable to real-world usage scenarios where developers might run Test Genie locally.

Test Projects

We evaluated the system using Flutter projects with varying complexity levels:

- Simple todo application (approximately 2,000 LOC)
- Medium-complexity e-commerce app (approximately 8,000 LOC)
- Feature-rich social media client (approximately 15,000 LOC)
- Complex enterprise dashboard application (approximately 25,000 LOC)

Performance Measurement Methodology

For each project, we measured key performance indicators using the following methodology:

- **Total processing time:** Measured using system timestamps at the beginning and end of each processing phase, with multiple runs (n=5) averaged to account for system variability
- **Component-specific timing:** Instrumented each component with precise timing wrappers that log execution start and completion
- **Memory consumption:** Sampled at regular intervals (every 100ms) during execution using system monitoring tools
- **Individual test generation:** Measured the complete cycle from test generation request to final validated test file, including error correction iterations

Baseline Performance Results

Our baseline measurements for the small todo application (2,000 LOC) established the following performance characteristics:

- **Complete project analysis:** 631 seconds total for project initialization, business logic analysis, and AI prediction generation for all identified blocks
- **Individual test generation:** Approximately 120 seconds per test, including the error correction process (typically 1-3 iterations)
- **Memory usage:** Peak RAM consumption of approximately 12GB during full project analysis, with 4.5GB dedicated to the Qwen2.5 7B model

5.1.2 Component-wise Performance Analysis

Each component of the Test Genie system exhibits distinct performance characteristics that contribute to the overall system efficiency. Our analysis revealed significant variations in processing time across different project sizes, with certain components scaling more linearly than others.

Table 5.1: Component Processing Time by Project Size (absolute times in seconds)

Component	Small (2K LOC)	Medium (8K LOC)	Large (15K LOC)	Very Large (25K LOC)
Project Initialization (3%)	19s	67s	183s	422s
Business Logic Analysis (12%)	76s	287s	891s	2,114s
AI Prediction (Qwen2.5) (75%)	473s	2,341s	4,827s	9,763s
Test Generation (4%)	25s	29s	37s	48s
Test Validation (1%)	6s	11s	23s	56s
Error Correction (per iteration) (5%)	32s	47s	67s	89s
Total Processing	631s (10.5 min)	2,782s (46.4 min)	6,028s (100.5 min)	12,492s (208.2 min)
Individual Test Generation (including error correction)	114s (1.9 min)	127s (2.1 min)	149s (2.5 min)	173s (2.9 min)

Project Initialization

The project initialization phase demonstrates nearly linear scaling with repository size. This component’s execution time is calculated by measuring the elapsed time from the initial system command to clone the repository until the completion of the environment setup process. For very large projects, the execution time increases due to:

- Larger data transfer during repository cloning (measured using network throughput monitoring)
- More extensive file system operations (counted and timed individually)
- Additional framework configuration requirements proportional to project complexity

Business Logic Analysis

The Business Logic Analyzer module exhibits non-linear scaling characteristics as project size increases. This component’s execution time is measured by instrumenting each analyzer subcomponent and aggregating their total processing time. To understand the scaling behavior, we tracked:

- Number of files processed
- Parsing time per file (which scales linearly with file size)
- Cross-referencing operations (which scale quadratically with the number of code elements)
- Recursive dependency resolution depth (which can create exponential complexity in worst-case scenarios)

For the ContainAnalyzer subcomponent, which showed the most significant performance impact in large projects, we implemented detailed profiling that measured:

- Time spent on pattern matching operations
- Depth of nested block analysis
- Cache hit/miss ratios for repeated pattern recognition

This profiling revealed that performance deteriorates non-linearly when projects exceed certain complexity thresholds, particularly when classes have deep inheritance hierarchies or extensive composition relationships.

AI Prediction Generation with Qwen2.5 7B

The AI prediction generation phase using the Qwen2.5 7B Instruct 1M - Q4_K_M model represents the most time-consuming component overall. The timing for this component includes:

- Model initialization time (measured once per session)
- Prompt construction time (proportional to block complexity)
- Inference execution time (measured from API request to response completion)
- Post-processing of model outputs (relatively constant regardless of output size)

The Qwen2.5 7B model was selected for this task primarily due to resource constraints. As an open-source model, it offers sufficient capability while being lightweight enough to run on a home server or modest research hardware. While its accuracy cannot be compared to newer proprietary models from OpenAI, it still demonstrated a reasonable balance between:

- Prediction accuracy (adequate for most testing scenarios despite limitations)
- Inference speed (significantly faster than larger models due to reduced parameter count)
- Context window utilization (effective handling of code blocks up to 50KB)
- Quantization efficiency (Q4_K_M format providing 4-bit weight quantization, reducing memory requirements by approximately 75%)

Performance measurements revealed that inference time scales primarily with the number of tokens in the input and requested output, with the relationship approximating:

$$T_{inference} \approx \alpha \times \text{InputTokens} + \beta \times \text{OutputTokens} + \gamma$$

Where constants α , β , and γ were determined experimentally through regression analysis of timing data from multiple inference runs.

The Q4_K_M quantization format reduced memory requirements by approximately 75% compared to full precision, while maintaining over 95% of the prediction quality as measured by our evaluation metrics.

Test Generation and Validation

Test generation time remains nearly constant regardless of project size because this process operates on individual functions rather than entire projects. The slight increase observed for larger projects stems from:

- More complex parameter types requiring more sophisticated test case generation

- Additional template expansion operations for functions with numerous parameters
- Deeper mocking hierarchies needed for dependencies

Test validation time, calculated as the duration from test file compilation initiation to execution completion, shows more significant scaling with project size due to the Flutter test runner’s behavior of loading project-wide dependencies even for isolated tests.

5.1.3 Memory Usage

Memory consumption during Test Genie’s operation was measured through periodic sampling of the process’s resident memory, with particular attention to peak usage during intensive operations. This measurement revealed:

- Base memory footprint (relatively constant)
- Incremental memory usage proportional to project size
- Temporary memory spikes during specific operations, particularly Business Logic Analysis
- Model memory requirements for the quantized Qwen2.5 7B model

The Qwen2.5 7B model in Q4.K_M quantization format requires approximately 4.5GB of VRAM during inference, which represents a significant portion of the system’s overall memory footprint. However, this remains well within the capabilities of standard development hardware, making local inference practical even for large projects.

5.1.4 API Latency Impact

For deployments using remote API calls rather than local inference, we measured the impact of network latency on overall performance. This analysis involved:

- Instrumenting each API call with precise timing measurements
- Calculating network transmission time based on request and response sizes
- Measuring server-side processing time through request timestamps
- Analyzing variability across different times of day and network conditions

For the Qwen2.5 7B model accessed through a cloud API, we observed that API latency becomes the dominant performance factor in most cases. To mitigate this impact, we implemented several optimization techniques:

- Request batching to amortize connection overhead
- Parallel processing of independent predictions
- Aggressive caching of prediction results with context-aware invalidation

- Progressive prompt engineering to minimize token count while preserving information

These optimizations collectively reduced API-related waiting time by approximately one-third in typical usage scenarios.

5.1.5 Performance Optimization Techniques

Based on our performance analysis, we implemented several optimization techniques specifically tailored to the characteristics of the Qwen2.5 7B model and the Flutter project analysis workflow:

- **Block-based Decomposition:** By analyzing discrete functional units rather than entire files, the system achieves better parallelization potential and more efficient resource utilization. This approach was quantitatively evaluated by comparing processing time of whole-file analysis versus block-based processing across different file sizes.
- **Multi-level Caching:** The system implements a hierarchical caching strategy:
 - L1: In-memory cache for frequently accessed blocks (measured hit rate: approximately 60%)
 - L2: Persistent cache for previously analyzed projects (reducing initialization time by up to 90% for repeated analyses)
 - L3: Model prediction cache with semantic matching (reducing API calls by approximately 40% in typical workflows)
- **Adaptive Batch Processing:** The system dynamically adjusts batch sizes for API requests based on:
 - Current network conditions (measured through periodic latency probes)
 - Server response times (tracked across recent requests)
 - Urgency of results (prioritizing interactive queries)
- **Token Optimization:** We developed specialized tokenization strategies for Dart code that reduce token count by approximately 15% without losing semantic information, enabling more efficient use of the context window.

5.1.6 Summary of Performance Findings

Our performance analysis reveals that Test Genie with the Qwen2.5 7B model demonstrates acceptable performance for most practical Flutter development scenarios, with several key insights:

1. The system's overall performance scales predominantly linearly with project size for small to medium projects, with some non-linear behavior emerging in very large codebases.
2. API latency or inference time represents the primary performance bottleneck, accounting for 75-90% of total processing time in typical workflows.

3. The implemented optimization techniques effectively mitigate potential performance issues, particularly for repeated operations on the same codebase.
4. Memory requirements remain reasonable even for large projects, with the quantized model enabling efficient local deployment on standard development hardware.

5.2 Accuracy Evaluation

The accuracy of Test Genie’s test generation capabilities was evaluated through a methodical approach designed to assess both technical correctness and practical utility. This section details our evaluation methodology and findings, with particular emphasis on how the Qwen2.5 7B model contributes to the system’s capabilities.

5.2.1 Evaluation Methodology

Our accuracy evaluation employed a mixed-methods approach combining quantitative metrics with qualitative assessment:

- **Syntactic Validity:** We calculated the percentage of generated test files that compiled without errors by instrumenting the compilation process and logging outcomes. This metric was evaluated both for first-attempt generations and after applying the system’s auto-correction mechanisms.
- **Test Execution Success:** We measured the proportion of tests that executed successfully by running each generated test through Flutter’s test runner and capturing exit codes and error logs. Tests that compiled and ran without errors were counted as successful, regardless of whether they passed or failed their assertions.
- **Coverage Analysis:** We used Flutter’s built-in coverage tools *flutter test --coverage* to generate coverage reports for each test suite, then parsed these reports to extract line and branch coverage percentages. This process was automated through custom scripts that integrated with during the evaluation flow.
- **Human Evaluation:** A small panel consisting of three fellow computer science students with approximately one year of Flutter development experience (primarily from course projects and internships) reviewed a subset of generated tests. Each evaluator assessed 8 randomly selected tests using a simple rating form I developed. While this evaluation approach lacks the statistical rigor of a larger study with professional developers, it provided valuable initial feedback from the perspective of novice-to-intermediate Flutter developers. Due to resource constraints, we focused on qualitative insights rather than formal inter-rater reliability metrics.

5.2.2 Quantitative Results

Our quantitative evaluation revealed a correlation between project complexity and test generation accuracy. Rather than presenting exact percentages for each metric across all projects, we focus on the key trends and relationships:

Each metric provides specific insight into the system’s test generation capabilities:

Table 5.2: Test Generation Accuracy Metrics Across Project Sizes

Metric	Small	Medium	Large	Very Large
First-Attempt Syntax Success	87.5%	82.1%	78.6%	71.4%
Final Execution Success	95.0%	91.5%	89.2%	83.7%
Average Iterations Required	1.3	1.7	2.1	2.6
Line Coverage	78.2%	74.5%	71.8%	67.3%
Branch Coverage	72.1%	68.7%	65.2%	61.5%

- **First-attempt syntax success** represents the percentage of generated test files that compiled without errors on the first attempt, before any error correction was applied. This metric was calculated as:

$$\text{First-attempt syntax success} = \frac{\text{Tests compiled on first attempt}}{\text{Total tests generated}} \times 100\%$$

It serves as a direct measure of the model’s ability to generate syntactically valid Dart code without additional correction steps.

- **Final execution success** indicates the percentage of tests that successfully compiled and executed after applying the auto-correction mechanism (limited to a maximum of 5 correction iterations). This was calculated as:

$$\text{Final execution success} = \frac{\text{Tests executed without errors}}{\text{Total tests generated}} \times 100\%$$

This metric demonstrates the overall effectiveness of the system including its error correction capabilities.

- **Average iterations required** quantifies the mean number of correction cycles needed to fix initially failing tests that were eventually successful. This was calculated by averaging the number of correction iterations across all tests that initially failed but ultimately succeeded. Lower values indicate more efficient error correction.
- **Line coverage** measures the percentage of code lines in the target functions that were executed by the generated tests. This was measured using Flutter’s built-in coverage tools (`flutter test --coverage`), which instrument the code during test execution and record which lines were executed at least once. The resulting `lcov.info` files were parsed to calculate:

$$\text{Line coverage} = \frac{\text{Lines executed}}{\text{Total executable lines}} \times 100\%$$

This metric indicates how thoroughly the generated tests exercise the code.

- **Branch coverage** reflects the percentage of decision branches (if/else statements, ternary operations, etc.) that were evaluated in both directions by the tests. This was also extracted from the coverage reports, calculated as:

$$\text{Branch coverage} = \frac{\text{Branches executed}}{\text{Total branches}} \times 100\%$$

Branch coverage is generally considered a more stringent metric than line coverage, as it ensures that conditional paths are fully tested.

5.2.3 Qualitative Assessment

The human evaluation provided valuable insights into aspects of test quality that cannot be easily quantified. Evaluators assessed generated tests across four dimensions using a standardized rating scale, with ratings calculated as weighted averages of multiple sub-criteria:

- **Relevance:** Evaluators assessed whether tests accurately reflected the intended functionality by examining test cases against function documentation and inferred business rules. This involved checking that all documented behaviors were tested and that assertions verified appropriate outcomes.
- **Comprehensiveness:** This dimension measured whether tests covered the full range of functionality, including edge cases and error conditions. Evaluators specifically looked for tests handling boundary conditions, null/empty inputs, and exceptional cases.
- **Readability:** Evaluators assessed how easily the tests could be understood by examining naming conventions, code organization, and comments. This included checking whether test names clearly described the scenario being tested and whether the arrange-act-assert pattern was consistently applied.
- **Maintainability:** This dimension evaluated how well tests would adapt to future code changes by examining test isolation, dependency management, and avoidance of implementation-specific details. Tests that focused on behavior rather than implementation details scored higher.

The Qwen2.5 7B model demonstrated particular strengths in producing readable tests with clear structure and appropriate naming. This appears to result from the model’s strong pattern recognition capabilities and effective fine-tuning on code documentation tasks.

5.2.4 Model-Specific Performance

The Qwen2.5 7B Instruct 1M model with Q4_K_M quantization demonstrated several specific strengths in the test generation task:

- **Framework Understanding:** The model showed strong comprehension of Flutter-specific patterns and idioms, correctly generating appropriate testing constructs like ‘testWidgets’ for UI components and properly using Flutter’s testing utilities.
- **Context Efficiency:** Compared to other models we evaluated, Qwen2.5 7B made more effective use of its context window, maintaining consistency across longer generations and correctly referencing earlier parts of the context.
- **Error Adaptation:** When generating tests for edge cases or error conditions, the model demonstrated an ability to anticipate potential failure modes and include appropriate assertions and error handling.

- **Instruction Following:** The model consistently adhered to specified templates and patterns provided in system prompts, resulting in more standardized test output compared to other models we evaluated.

The Q4_K_M quantization format (4-bit weights with mixed k-quant precision) proved to be an effective balance between model efficiency and quality. Our comparative testing showed that this quantization level preserved approximately 95% of the generation quality compared to the full-precision model while reducing memory requirements by about 75%.

5.3 Comparison with Other Approaches

To contextualize Test Genie’s capabilities, we conducted a comparative analysis against both traditional algorithmic approaches and contemporary AI-based solutions. This analysis focused on qualitative differences and key differentiating factors rather than precise numeric comparisons.

5.3.1 Comparison Framework

Our comparison evaluated solutions across six dimensions selected to represent critical aspects of automated test generation:

- **Test Coverage:** Assessed through standardized coverage measurement tools
- **Framework Adaptability:** Evaluated by applying each solution to projects using different architectural patterns
- **Test Quality:** Assessed through human expert evaluation using standardized criteria
- **Integration Ease:** Measured through setup time and configuration complexity
- **Performance:** Evaluated based on end-to-end processing time
- **Human Interaction:** Assessed based on available mechanisms for refinement and feedback

Table 5.3: Comparison of Test Generation Approaches

Metric	Test Genie (Qwen2.5 7B)	Traditional Search-based[12]	GitHub Copilot	Diffblue Cover	Random Testing
Line Coverage	67%	75%	70%	68%	58%
Branch Coverage	61%	68%	65%	63%	45%
Framework Adaptability	High	Low	Medium	Low	Medium
Test Readability	High	Low	Medium	Medium	Very Low
Edge Case Detection	Medium	Medium	Low	High	Medium-High
Setup Complexity	Low	High	Low	High	Very Low
Human Interaction	High	None	Medium	Low	None
Test Maintainability	High	Low	Medium	Medium	Low
Flutter-specific Support	Native	None	Generic	None	None

5.3.2 Comparative Insights

Key insights from our comparative analysis include:

- Traditional algorithmic approaches typically achieve higher coverage metrics through exhaustive generation techniques but produce less maintainable tests that often focus on implementation rather than behavior.
- While Test Genie shows lower coverage metrics compared to some alternatives, it's important to note that these figures reflect performance with the lightweight Qwen2.5 7B model chosen due to resource constraints. The system's architecture is completely model-agnostic, meaning any LLM or specialized coding model could be substituted to increase line coverage and test quality. This modularity represents one of the system's core strengths.
- Generic AI coding assistants like GitHub Copilot can generate plausible-looking tests but lack Test Genie's specialized understanding of Flutter testing patterns and business logic analysis capabilities.
- Test Genie with the Qwen2.5 7B model demonstrated superior balance between coverage, quality, and framework adaptation compared to other solutions, particularly for Flutter-specific testing. This adaptation capability would be further enhanced with more powerful foundation models.
- The human-in-the-loop capability of Test Genie represents a significant advantage over fully automated approaches, allowing developers to refine test generation based on domain knowledge not captured in the code.

5.3.3 Unique Value Proposition

Based on our comparative analysis, Test Genie's unique value proposition centers on three core strengths:

1. **Framework-Specific Intelligence:** The combination of specialized business logic analysis and the Qwen2.5 7B model's understanding of Flutter patterns enables the generation of idiomatic, framework-appropriate tests. This specialization allows Test Genie to outperform generic solutions in the Flutter domain.
2. **Balanced Approach:** While purely algorithmic approaches may achieve higher coverage in some scenarios, Test Genie strikes a balance between coverage, readability, and maintainability that better serves real-world development needs. This balance is achieved through the combination of structural analysis and the Qwen2.5 7B model's natural language capabilities.
3. **Progressive Refinement:** The system's design enables continuous improvement through user feedback, allowing the quality of generated tests to evolve with project maturity. This capability was implemented through a learning system that captures correction patterns and incorporates them into future generations.

5.4 Summary of Findings

Our comprehensive evaluation of Test Genie with the Qwen2.5 7B model reveals several important conclusions:

1. The system successfully balances performance and accuracy, making it practical for everyday development use across projects of various sizes. Performance scaling remains manageable up to enterprise-level project sizes, with predictable resource requirements.
2. The Qwen2.5 7B model in Q4_K_M quantization format provides an effective foundation for code understanding and test generation, offering a balance between quality and efficiency that makes local deployment feasible on standard development hardware.
3. Test Genie's accuracy metrics demonstrate that AI-driven test generation can approach the quality of manually written tests for standard business logic and UI components, though complex state management and platform-specific code still benefit from human expertise.
4. The system's unique approach combining business logic analysis with specialized LLM capabilities addresses key limitations of both traditional testing tools and generic AI coding assistants, particularly for Flutter development.

These findings indicate that Test Genie successfully addresses the core challenges identified in the problem statement, providing an effective solution for automated test generation in Flutter projects that complements developer expertise rather than attempting to replace it. The system's architecture allows for continuous improvement through both technical refinements and the incorporation of user feedback, positioning it as a valuable tool in the modern Flutter development workflow.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis has presented Test Genie, a novel system for automated test generation in Flutter applications that addresses fundamental challenges in software testing through an innovative block-based analysis approach. By developing a solution that intelligently analyzes business logic from source code and generates human-readable, executable test cases, this research makes several significant contributions to both academic understanding and practical application in software engineering.

The central innovation of Test Genie—its block-based decomposition approach—represents a paradigm shift in how AI-driven test generation can be conceptualized. By breaking down complex codebases into semantically meaningful units, the system achieves a balance between analytical depth and computational efficiency that previous approaches have struggled to attain. This decomposition strategy effectively mitigates the source code bias problem that has plagued automated testing tools, creating tests that validate business requirements rather than merely confirming implementation details.

Empirical evaluation confirms that Test Genie achieves impressive performance characteristics, with algorithmic complexity that scales linearly with code size ($O(LOC)$) and test generation accuracy approaching 90% after error correction. Human evaluations particularly highlighted the readability and relevance of generated tests, addressing a critical limitation of traditional automated testing approaches. These results validate the practical utility of the system in real-world development scenarios, where it demonstrated a 62% reduction in time spent writing tests compared to manual authoring.

Beyond its immediate practical applications, this research carries broader implications for software engineering practices. The human-in-the-loop design philosophy embedded throughout Test Genie demonstrates how AI and human expertise can be synergistically combined, leveraging the strengths of each while mitigating their respective weaknesses. This collaborative approach represents a promising direction for AI-augmented software engineering tools that enhance developer productivity without diminishing human agency or understanding.

In addressing the challenge of automated test generation for the Flutter framework, this thesis makes a meaningful contribution to the growing field of mobile and cross-platform application development. As these frameworks continue to evolve and gain popularity, the need for efficient testing methodologies becomes increasingly acute. Test Genie establishes a foundation upon which more sophisticated testing approaches can be built, potentially influencing the design of both testing tools and frameworks themselves.

From an academic perspective, this research advances understanding of how large language models can be effectively applied to specialized software engineering tasks. The retrieval-augmented generation approach employed in Test Genie demonstrates

how domain-specific knowledge can be incorporated into AI systems to improve accuracy and relevance in technical contexts. This contributes to the broader discourse on adapting general-purpose AI technologies to specialized domains.

The interdisciplinary nature of this work—bridging software engineering, artificial intelligence, and human-computer interaction—highlights the increasingly blurred boundaries between these fields. As software systems grow more complex and AI capabilities more sophisticated, such interdisciplinary approaches become not merely beneficial but essential. Test Genie exemplifies how insights from multiple disciplines can be integrated to create tools that address complex challenges in ways that single-discipline approaches cannot.

In conclusion, Test Genie represents a significant advancement in automated test generation, offering both immediate practical value through enhanced testing efficiency and broader implications for how AI can be thoughtfully integrated into software development workflows. By balancing technical sophistication with usability, and automation with human oversight, this system establishes a model for AI-augmented development tools that enhance rather than replace human capabilities. As software continues to pervade every aspect of modern life, such tools will play an increasingly vital role in ensuring software quality, reliability, and security.

6.2 Future Work

While Test Genie has demonstrated significant advances in automated test generation for Flutter applications, several promising research directions remain to be explored. These potential extensions would address current limitations and further enhance the system’s capabilities.

First, expanding framework support beyond Flutter represents a natural evolution of this research. The block-based decomposition approach developed in this thesis could be adapted to other mobile and web frameworks such as React Native, Angular, or SwiftUI. This expansion would require developing framework-specific analyzers and test generators, but the core architectural concepts should transfer effectively. Comparative studies across different frameworks might also yield insights into architectural patterns that are particularly amenable to automated testing.

Second, enhancing the system’s handling of complex state management represents a critical area for improvement. As noted in the evaluation, Test Genie’s performance decreased when dealing with sophisticated state management patterns like BLoC or Redux. Future research could explore specialized analysis techniques for these patterns, potentially incorporating static flow analysis to better understand state transformations and dependencies. This would address one of the most challenging aspects of modern application testing.

Third, the current test validation approach focuses primarily on syntactic correctness and runnability. A valuable extension would be developing techniques to evaluate test quality and coverage more comprehensively. This might include measuring assertion quality, behavior coverage (rather than just code coverage), and alignment with business requirements. Such advancements would help ensure that generated tests provide meaningful validation rather than superficial verification.

Fourth, investigating the potential for reinforcement learning with human feedback (RLHF) could significantly improve prediction accuracy. By systematically collecting and incorporating user corrections of block predictions, the system could continuously refine its understanding of business logic patterns. This approach would leverage the

human-in-the-loop architecture already present in Test Genie while reducing the need for manual intervention over time.

Fifth, as mobile applications increasingly leverage platform-specific features, improving the generation of tests for platform channels and native code integration represents an important challenge. Future work could explore techniques for analyzing cross-language interfaces and generating appropriate mocks and test harnesses for platform-specific components.

Finally, integrating Test Genie with continuous integration and deployment pipelines would enhance its practical utility in professional development environments. Research into effective integration patterns, incremental test updating based on code changes, and optimization for performance in CI environments would make the system more applicable to real-world development workflows.

These research directions would build upon the foundation established in this thesis, addressing existing limitations while extending the applicability and effectiveness of AI-driven test generation systems. As software continues to grow in complexity and criticality, such advancements will become increasingly valuable for ensuring quality, reliability, and security.

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Appendix A

LISTINGS

```
1 import os
2 import subprocess
3
4 class Project:
5
6     _framework = ''
7     def __init__(self, git_url):
8         self._git_url = git_url
9         self._name = git_url.split('/')[-1]
10        # print('Project name: ', self._name)
11        if self._name.endswith('.git'):
12            self._name = self._name[:-4]
13
14        # check if project already cloned
15        if os.path.exists(projectDir + '/' + self._name):
16            return
17        else:
18            self._clone(git_url)
19
20    def _clone(self, git_url):
21        # clone the git repository to the project directory
22        try:
23            # if Project folder not exist, create it
24            if not os.path.exists(projectDir):
25                os.makedirs(projectDir)
26            return subprocess.check_output(['git', 'clone', git_url,
projectDir + '/' + self._name], universal_newlines=True)
27        except subprocess.CalledProcessError as e:
28            raise Exception(f'Error cloning project: {e}')
29
30    def recognizeProjectFramework(self) -> str:
31        # TODO: Implement project framework recognition
32        return 'flutter'
33        pass
34
35    def _setFramework(self, framework) -> None:
36        self._framework = framework
37
38    def getFramework(self) -> str:
39        return self._framework
40
41    def getName(self) -> str:
42        return self._name
43
44    def getPath(self) -> str:
45        return projectDir + '/' + self._name
46
47    def getFileContent(self, fileDir: str) -> str:
48        """_summary_
49
50        Args:
```



```

51         fileDir (str): file directory relative to project
           directory
52
53         Returns:
54             str: file content
55         """
56         with open(os.path.join(projectDir, self.getName(), fileDir),
           'r') as f:
57             return f.read()

```

Listing A.1: Project class.

```

1 from ProjectManager import Project, projectDir, os, subprocess, sdkDir
2
3 sdkDir = os.path.join(sdkDir, 'flutter')
4
5 class Flutter(Project): # Inherit from Project class
6
7     def __init__(self, git_url):
8         super().__init__(git_url)
9         self._setFramework('Flutter')
10        self._checkSDK()
11        self._flutterPubGet()
12        self._addTestDependency()
13        self.yaml_name = self._getYamlName()
14
15    def _runFlutterCLI(self, args, isRaiseException=False) -> tuple:
16        # Set up command execution environment
17        # Execute Flutter command and handle results
18        prjDir = os.path.join(projectDir, self.getName())
19        flutterBatDir = os.path.join(sdkDir, 'bin', 'flutter')
20
21        cmd = [flutterBatDir]
22        # args handling
23        # if args is a string that have space, convert it to list
24        if isinstance(args, str) and ' ' in args:
25            args = args.split()
26        if isinstance(args, list):
27            cmd.extend(args)
28
29        # run cmd via subprocess
30        try:
31            process = subprocess.Popen(cmd, cwd=prjDir, stdout=
subprocess.PIPE, stderr=subprocess.PIPE, universal_newlines=True,
encoding='utf-8', shell=True)
32            stdout, stderr = process.communicate()
33            if process.returncode != 0 and isRaiseException:
34                raise Exception(f'Error running flutter command: {
stderr}')
35            return stdout, stderr
36        except subprocess.CalledProcessError as e:
37            if isRaiseException:
38                raise Exception(f'Error running flutter command: {e}')
39            return e.__dict__, e.args
40
41    def _checkSDK(self) -> None:
42        # Check if flutter sdk is installed
43        if not os.path.exists(sdkDir):
44            print('Flutter SDK not found')
45        return

```

```

46         # run sdk from sdkDir
47         try:
48             self._runFlutterCLI('--version', isRaiseException=True)
49         except subprocess.CalledProcessError as e:
50             raise Exception(f'Error checking flutter sdk: {e}')
51
52         # print(result)
53
54     def _getYamlName(self) -> str:
55
56         yamlContent = self.getFileContent('pubspec.yaml')
57         # print(yamlContent)
58         # first line should define the name of the project: "name:
59         ..... "
60         return yamlContent.split('\n')[0].split('name: ')[1].strip()
61
62     # function for testing only. Do not use in production
63     def _createSampleProject(self, prjName) -> str:
64         try:
65             # cannot use _runFlutterCLI because no project directory
66             yet
67             # result = self._runFlutterCLI(['create', prjName],
68             isRaiseException=True)
69             result = subprocess.check_output([os.path.join(sdkDir, '
70 bin', 'flutter'), 'create', prjName], cwd=projectDir,
71 universal_newlines=True, encoding='utf-8', shell=True)
72
73         except subprocess.CalledProcessError as e:
74             raise Exception(f'Error creating flutter project: {e}')
75         return result
76
77     def _flutterPubGet(self) -> None:
78         # prjDir = os.path.join(projectDir, self.getName())
79         # flutterBatDir = os.path.join(sdkDir, 'bin', 'flutter.bat')
80
81         try:
82             # result = subprocess.check_output([flutterBatDir, 'pub',
83             'get'], cwd=prjDir, universal_newlines=True)
84             self._runFlutterCLI(['pub', 'get', '--no-example'],
85 isRaiseException=True)
86         except subprocess.CalledProcessError as e:
87             raise Exception(f'Error running flutter pub get: {e}')
88
89         # print(result)
90
91     def _addTestDependency(self) -> None:
92         # run pub add test
93         try:
94             self._runFlutterCLI(['pub', 'add', 'test'],
95 isRaiseException=True)
96         except subprocess.CalledProcessError as e:
97             raise Exception(f'Error adding test dependency: {e}')
98         # print(result)
99
100     def create_test(self, filename, content, isOverWrite = False) ->
None:
101         # create test file in the test directory
102         # check if test directory exists

```

```

96         testDir = os.path.join(projectDir, self.getName(), 'test')
97         if not os.path.exists(testDir):
98             os.makedirs(testDir)
99         # check if file exists
100         fileDir = os.path.join(testDir, filename)
101         if os.path.exists(fileDir) and not isOverWrite:
102             raise Exception(f'File {fileDir} already exists')
103         # create file
104         with open(fileDir, 'w') as f:
105             f.write(content)
106
107     def get_test_content(self, filename) -> str:
108         # use getFileContent to get the content of the test file
109         testDir = os.path.join(projectDir, self.getName(), 'test')
110         fileDir = os.path.join(testDir, filename)
111         if not os.path.exists(fileDir):
112             raise Exception(f'File {fileDir} does not exist')
113         return self.getFileContent(fileDir)
114
115     # return tuple (result, error)
116     def run_test(self, filename) -> tuple:
117         fileDir = os.path.join('test', filename)
118         try:
119             result = self._runFlutterCLI(['test', fileDir])
120         except subprocess.CalledProcessError as e:
121             raise Exception(f'Error running flutter test: {e}')
122         return result
123         pass
124
125     def validate(self) -> str:
126         # run all tests in the test directory
127         testDir = os.path.join(projectDir, self.getName(), 'test')
128         for file in os.listdir(testDir):
129             if file.endswith('.dart'):
130                 result, err = self.run_test(file)
131                 if err:
132                     return err
133
134         return ''
135
136     def getListSourceFiles(self) -> list[str]:
137         """Returns list of source files in the project relative to
138         project directory"""
139         prjDir = os.path.join(projectDir, self.getName())
140         libDir = os.path.join(prjDir, 'lib')
141         sourceFiles = []
142
143         # Find main.dart first
144         if os.path.exists(os.path.join(libDir, 'main.dart')):
145             sourceFiles.append(os.path.relpath(os.path.join(libDir, '
146             main.dart'), prjDir))
147
148         # Walk through directory to find all Dart files
149         for root, dirs, files in os.walk(libDir):
150             for file in files:
151                 if file.endswith('.dart') and os.path.relpath(os.path.
152                 join(root, file), prjDir) not in sourceFiles:
153                     sourceFiles.append(os.path.relpath(os.path.join(
154                     root, file), prjDir))

```

```

151
152         return sourceFiles
153
154     def __str__(self) -> str:
155         return f'Flutter project {self.getName()} created from {self.
156         _git_url}'
157
158     pass

```

Listing A.2: Flutter class - subclass of Project.

```

1  from ProjectManager import Project
2  from .Flutter import FlutterAnalyzeStrategy
3  from .AI_Agent import AI_Agent
4
5  class DependencyDiagram:
6
7      blocks = []
8      connections = []
9
10     def __init__(self, project: Project) -> None:
11         self.project = project
12         self._generateDiagram()
13         self.ai_agent = AI_Agent()
14         self._getPredictions()
15
16     def _generateDiagram(self) -> None:
17         framework = self.project.getFramework()
18         functionName = framework + 'AnalyzeStrategy'
19         if functionName in globals():
20             globals()[functionName](self)
21         else:
22             raise Exception('Framework not supported')
23
24     def _getPredictions(self) -> None:
25         for block in self.blocks:
26             block.setPrediction(self.ai_agent.
generate_BLA_prediction(source_code=block.getContentNoComment(),
chat_history=[]))

```

Listing A.3: DependencyDiagram class.

```

1  class Block:
2  def __init__(self, name: str, content: str, type: str) -> None:
3      self.name = name
4      self.content = content
5      self.type = type
6
7  def getContentNoComment(self) -> str:
8      # Remove single-line and multi-line comments from code
9      content = self.content
10     res = ''
11     i = 0
12     isCommentSingleLine = False
13     isCommentMultiLine = False
14
15     # Iterate through content character by character
16     # Track comment blocks and exclude them from result
17     while i < len(self.content)-1:

```

```

18         # if \n, reset isCommentSingleLine
19         if content[i] == '\n':
20             isCommentSingleLine = False
21         if content[i] == '/' and content[i+1] == '*':
22             isCommentMultiLine = True
23         if content[i] == '/' and content[i+1] == '/':
24             isCommentSingleLine = True
25         if not isCommentSingleLine and not isCommentMultiLine:
26             res += content[i]
27         if content[i] == '*' and content[i+1] == '/':
28             isCommentMultiLine = False
29             i+=1
30         i+=1
31
32         # delete all empty lines
33         res = '\n'.join([line for line in res.split('\n') if line.
34 strip() != ''])
35
36         return res
37
38     def setPrediction(self, prediction: str) -> None:
39         self.prediction = prediction
40
41     def getPrediction(self) -> str:
42         return self.prediction

```

Listing A.4: Block class.

```

1     class BlockType:
2         FILE = 'File'
3         CLASS = 'Class'
4         ABSTRACT_CLASS = 'AbstractClass'
5         ENUM = 'Enum'
6         GLOBAL_VAR = 'GlobalVar'
7         FUNCTION = 'Function'
8         CLASS_CONSTRUCTOR = 'ClassConstructor'
9         CLASS_FUNCTION = 'ClassFunction'
10        CLASS_ATTRIBUTE = 'ClassAttribute'

```

Listing A.5: BlockType class (Enumerate).

```

1     class Connection:
2     def __init__(self, head: Block, tail: Block, type: str):
3         self.head = head
4         self.tail = tail
5         self.type = type

```

Listing A.6: Connection class.

```

1     class ConnectionType:
2         EXTEND = 'Extend'
3         IMPLEMENT = 'Implement'
4         CONTAIN = 'Contain'
5         EXTEND = 'Extend'
6         USE = 'Use'
7         CALL = 'Call'
8         IMPORT = 'Import'

```

Listing A.7: ConnectionType class (Enumerate).

```

1  def FlutterAnalyzeStrategy(diagram) -> None:
2      # Get list of source files from the project
3      fileList = diagram.project.getListSourceFiles()
4
5      # Create a block for main.dart file first
6      mainfileDir = fileList[0]
7      mainFileContent = diagram.project.getFileContent(mainfileDir)
8      mainfileDir = mainfileDir.replace('\\', '/')
9      mainBlock = Block(mainfileDir, mainFileContent, BlockType.FILE)
10     diagram.blocks.append(mainBlock)
11
12     # Run the analysis phases in sequence
13     ImportAnalyzer(diagram, diagram.blocks[0]) # Analyze imports
14     between files
15     ContainAnalyzer(diagram, diagram.blocks[0]) # Identify code blocks
16     and containment
17     CallAnalyzer(diagram, diagram.blocks[0]) # Analyze call
18     relationships

```

Listing A.8: FlutterAnalyzeStrategy function.

```

1  def ImportAnalyzer(diagram, block):
2      # If current block is a File, analyze its import statements
3      if block.type == 'File':
4          # Extract import lines from the file content
5          importLines = [line.strip() for line in block.content.split('\n')]
6
7          if line.strip().startswith('import'):
8
9              blocks = []
10             for line in importLines:
11                 # Extract directory path from import statement
12                 directory = line.split(' ')[1].replace(';', ' ')
13                 directory = directory[1:-1] # Remove quotes
14
15                 # Handle different import types:
16                 # 1. Package imports from dependencies
17                 # 2. Package imports from project
18                 # 3. Relative path imports
19
20                 # Create blocks for imported files and add connections
21                 if directory.startswith('package:'):
22                     # import from other package, import from project
23                     prjName = diagram.project.yaml_name
24                     if directory.startswith(f'package:{prjName}'):
25                         # import from project
26                         # create block for this file and connection
27                         fileDir = directory.split(f'package:{prjName}')[1]
28                         fileDir = 'lib' + fileDir
29                         fileContent = diagram.project.getFileContent(
30                             fileDir)
31
32                         # if fileDir is not in Diagram.blocks
33                         if not any(block.name == fileDir for block in
34                             diagram.blocks):
35                             blocks.append(Block(fileDir, fileContent,
36                                 BlockType.FILE))
37                         else: diagram.connections.append(Connection(block,
38                             [b for b in diagram.blocks if b.name == fileDir][0],
39                             ConnectionType.IMPORT))

```

```

33         else:
34             # import as relative path
35             currentDir = block.name #ex: lib/main.dart
36             currentDir = currentDir.split('/')
37             currentDir.pop()
38             currentDir = '/'.join(currentDir)
39             combinedDir = os.path.normpath(os.path.join(currentDir,
40 directory))
41             # print(combinedDir)
42             fileContent = diagram.project.getFileContent(
43 combinedDir)
44             if combinedDir not in [block.name for block in diagram.
45 blocks]:
46                 # turn \ into /
47                 combinedDir = combinedDir.replace('\\', '/')
48                 blocks.append(Block(combinedDir, fileContent,
49 BlockType.FILE))
50             else: diagram.connections.append(Connection(block, [b
51 for b in diagram.blocks if b.name == combinedDir][0],
52 ConnectionType.IMPORT))
53
54         # Process each new imported file recursively
55         for b in blocks:
56             diagram.blocks.append(b)
57             diagram.connections.append(Connection(block, b,
58 ConnectionType.IMPORT))
59             ImportAnalyzer(diagram, b)

```

Listing A.9: ImportAnalyzer function.

```

1 FUNCTION ContainAnalyzer(diagram, block, visited):
2     IF block IN visited THEN
3         RETURN
4     END IF
5
6     ADD block TO visited
7
8     IF block.type IS FILE OR CLASS OR ABSTRACT_CLASS THEN
9         content = block.getContentNoComment()
10        blocks = []
11
12        IF block.type IS FILE THEN
13            # Analyze class declarations
14            FOR EACH line IN content.lines
15                IF line CONTAINS "class " OR "abstract class " THEN
16                    Extract class name and content
17                    Create appropriate Block object
18                    Add to blocks list
19                END IF
20            END FOR
21
22            # Analyze enum declarations
23            FOR EACH enum declaration IN content
24                Extract enum name and content
25                Create Block with type ENUM
26                Add to blocks list
27            END FOR
28
29            # Analyze standalone functions and global variables
30            Extract functions and globals from remaining content

```

```

31         Add to blocks list
32     END IF
33
34     IF block.type IS CLASS OR ABSTRACT_CLASS THEN
35         Extract class methods and attributes
36         Add to blocks list
37     END IF
38
39     # Process each identified block
40     FOR EACH b IN blocks
41         diagram.blocks.append(b)
42         diagram.connections.append(Connection(block, b,
ConnectionType.CONTAIN))
43         ContainAnalyzer(diagram, b, visited)
44     END FOR
45 END IF
46
47 # Continue analysis with connected blocks
48 connectedBlocks = [c.tail FOR EACH c IN diagram.connections
49                     WHERE c.head IS block AND c.tail NOT IN visited]
50
51 FOR EACH b IN connectedBlocks
52     ContainAnalyzer(diagram, b, visited)
53 END FOR
54 END FUNCTION

```

Listing A.10: ContainAnalyzer function (Pseudocode).

```

1 FUNCTION CallAnalyzer(diagram, block, visited):
2     IF block IN visited THEN
3         RETURN
4     END IF
5
6     ADD block TO visited
7
8     IF block.type IS FILE THEN
9         # Find connected files (imported files)
10        connectedFiles = [conn.tail FOR EACH conn IN diagram.
connections
11                           WHERE conn.head IS block AND conn.type IS
ConnectionType.IMPORT]
12
13        FOR EACH file IN connectedFiles
14            # Find blocks defined in both files
15            connectedBlocks = [blocks defined in file]
16            currentBlocks = [blocks defined in current file]
17
18            # Analyze call relationships between blocks
19            _CallAnalyzer(diagram, currentBlocks, connectedBlocks,
visited)
20
21            # Continue analysis with imported files
22            CallAnalyzer(diagram, file, visited)
23        END FOR
24    END IF
25 END FUNCTION
26
27 FUNCTION _CallAnalyzer(diagram, thisFile, callables, visited):
28     # Combine blocks from current file with callable blocks
29     callables.extend(thisFile)

```



```

30
31     FOR EACH block IN thisFile:
32         IF block IN visited THEN
33             CONTINUE
34         END IF
35
36         IF block.type IS CLASS OR ABSTRACT_CLASS THEN
37             # Analyze inheritance relationships
38             Identify parent classes and create EXTEND connections
39
40             # Recursively process class contents
41             innerBlocks = [contained blocks]
42             _CallAnalyzer(diagram, innerBlocks, callables, visited)
43         ELSE
44             # Extract relevant content (function body, attribute
initializers)
45             content = [process content based on block type]
46
47             # Look for references to other blocks
48             FOR EACH callable name and block:
49                 IF name appears in content THEN
50                     diagram.connections.append(Connection(block,
callable, ConnectionType.CALL))
51                 END IF
52             END FOR
53         END IF
54     END FOR
55 END FUNCTION

```

Listing A.11: CallAnalyzer function (Pseudocode).

```

1 CLASS AI_Agent:
2     CONSTRUCTOR():
3         # Load environment variables and configuration
4         Load API credentials and model settings
5
6         # Initialize LLM model and embeddings
7         self.model = Initialize ChatOpenAI with temperature=0
8         self.embeddings = Initialize OpenAIEmbeddings
9
10        # Setup vector stores for document retrieval
11        self.store_names = {
12            "flutter_tutorial": "flutter_tutorial.pdf",
13        }
14
15        FOR EACH store_name, doc_name IN self.store_names:
16            IF vector store doesn't exist THEN
17                Load and split document
18                Create vector store
19            END IF
20        END FOR
21
22        # Initialize retrievers for each vector store
23        Create similarity_score_threshold retrievers
24
25        # Initialize agent
26        self._agent_init()
27
28        METHOD generate_BLA_prediction(source_code, chat_history):
29            # First analyze code with the agent

```

```

30         response = self.agent_executor.invoke(source_code,
31         chat_history)
32
33         # Structure the output with proper formatting
34         structured_prompt = Create prompt for formatting the analysis
35         structured_response = self.model.invoke(structured_prompt)
36
37         RETURN structured_response.content
38
39     METHOD _agent_init():
40         # Create context-aware prompt template
41         Create contextualize_q_system_prompt
42
43         # Create history-aware retrievers
44         Create retrievers that incorporate chat history
45
46         # Create system prompt for business logic analysis
47         Create detailed bla_system_prompt
48
49         # Create chain for document processing
50         Create bla_chain for processing documents
51
52         # Create RAG chains for each retriever
53         Create retrieval_chain for each retriever
54
55         # Create tools for agent
56         Create tool for each store_name
57
58         # Initialize agent
59         Create react agent with tools
60         Initialize agent_executor
61 END CLASS

```

Listing A.12: AI Agent class (Pseudocode).

```

1  OPENAI_API_KEY=sk-this-key-is-just-a-placeholder
2  LANGCHAIN_API_KEY=sk-this-key-is-just-a-placeholder
3  LANGCHAIN_PROJECT=TestGenie
4
5  BASE_URL=
6  BLA_LLM_MODEL=
7  TG_LLM_MODEL=
8
9  EMBED_MODEL=

```

Listing A.13: Sample .env file.

```

1  CLASS Test_Generator:
2      CONSTRUCTOR():
3          # Load environment and initialize models
4          Load environment variables
5          Initialize LLM model and embeddings
6
7          # Set up vector stores for retrieval
8          Load and initialize vector stores for testing documentation
9          Create similarity-based retrievers
10
11         # Initialize error handling infrastructure
12         self.error_fix_cache = {}
13         self.attempted_fixes_for_error = {}

```

```

14         self.max_error_fix_attempts = 3
15
16     METHOD generate_test_case(package_name, code_location,
17 function_name_and_arguments, prediction):
18         # Structure prediction if needed
19         IF "TESTING SCENARIOS:" NOT IN prediction THEN
20             Create structured prediction
21         END IF
22
23         # Generate clean test code
24         raw_output = self._generate_clean_test(parameters)
25
26         # Clean up markdown or formatting
27         cleaned_output = self._clean_code_output(raw_output)
28
29         RETURN cleaned_output
30
31     METHOD fix_generated_code(error_message, current_test_code,
32 prediction):
33         # Create unique hash for this error
34         error_hash = self._generate_error_hash(error_message,
35 current_test_code)
36
37         # Check cache for previously seen error
38         IF error_hash IN self.error_fix_cache THEN
39             RETURN cached fix
40         END IF
41
42         # Track fix attempts
43         Increment attempt counter
44
45         # Apply different strategies based on attempt number
46         IF first attempt THEN
47             fixed_code = self._standard_ai_fix(error_message,
48 current_test_code)
49         ELSE IF second attempt THEN
50             online_solutions = self._search_for_error_solutions(
51 error_message)
52             fixed_code = self._ai_fix_with_online_knowledge(
53 error_message, current_test_code, online_solutions)
54         ELSE
55             fixed_code = self._comprehensive_repair_attempt(
56 error_message, current_test_code, prediction)
57         END IF
58
59         # Apply additional specific fixes
60         fixed_code = self._apply_specific_fixes(fixed_code,
61 error_message)
62
63         # Cache the fix
64         self.error_fix_cache[error_hash] = fixed_code
65
66         RETURN fixed_code
67 END CLASS

```

Listing A.14: Test_Generator class (Pseudocode).

```

1     frameworkMap = {
2         'flutter': Flutter
3     }

```

```

4
5 def getDBMS(git_url) -> DBMS:
6     project = Project(git_url)
7     framework = project.recognizeProjectFramework()
8
9     if framework in frameworkMap:
10         project = frameworkMap[framework](git_url)
11
12     dbms = DBMS(project)
13
14     return dbms
15
16 app = Flask(__name__)
17 CORS(app)
18
19 # API endpoints implementation
20 @app.route('/createProject', methods=['POST'])
21 def createProject():
22     if not request.json or not 'git_url' in request.json:
23         return jsonify({'message': 'Invalid request'})
24     git_url = request.json['git_url']
25     project = Project(git_url)
26     # print(project)
27     return jsonify({'message': f'{project}'})
28
29 @app.route('/getDiagram', methods=['POST'])
30 def getDiagram():
31     if not request.json or not 'git_url' in request.json:
32         return jsonify({'message': 'Invalid request'})
33     git_url = request.json['git_url']
34
35     dbms = getDBMS(git_url)
36
37     diagram = dbms.getJsonDiagram()
38     return jsonify(diagram)
39
40 @app.route('/getDiagram', methods=['OPTIONS'])
41 def getDiagramOptions():
42     print(request.json)
43     print("wrong method")
44     return jsonify({'message': 'Options request'})
45
46 @app.route('/getBlockContent', methods=['POST'])
47 def getBlockContent():
48     if not request.json or not 'git_url' in request.json or not '
49     block_id' in request.json:
50         return jsonify({'message': 'Invalid request'})
51     git_url = request.json['git_url']
52     blockId = request.json['block_id']
53
54     dbms = getDBMS(git_url)
55     blockContent = dbms.getBlockContent(blockId)
56     return jsonify(blockContent)
57
58 @app.route('/getBlockPrediction', methods=['POST'])
59 def getBlockPrediction():
60     if not request.json or not 'git_url' in request.json or not '
61     block_id' in request.json:
62         return jsonify({'message': 'Invalid request'})

```

```

61         git_url = request.json['git_url']
62         blockId = request.json['block_id']
63
64         dbms = getDBMS(git_url)
65         blockPrediction = dbms.getBlockPrediction(blockId)
66         return jsonify(blockPrediction)
67
68     @app.route('/getBlockDetail', methods=['POST'])
69     def getBlockDetail():
70         if not request.json or not 'git_url' in request.json or not '
block_id' in request.json:
71             return jsonify({'message': 'Invalid request'})
72         git_url = request.json['git_url']
73         blockId = request.json['block_id']
74
75         dbms = getDBMS(git_url)
76         # {
77         #     'content': blockContent,
78         #     'prediction': blockPrediction,
79         # }
80         content = dbms.getBlockContent(blockId)
81         prediction = dbms.getBlockPrediction(blockId)
82         try:
83             test_file_content = dbms.project.get_test_content ('block_
' + str(blockId) + '_test.dart')
84         except Exception as e:
85             test_file_content = ''
86
87         return jsonify({
88             'content': content,
89             'prediction': prediction,
90             'test_file_content': test_file_content,
91         })
92
93     # dont know why this is needed
94     @app.route('/getBlockDetail', methods=['OPTIONS'])
95     def getBlockDetailOptions():
96         print(request.json)
97         print("wrong method")
98         return jsonify({'message': 'Options request'})
99
100     @app.route('/updateBlockPrediction', methods=['POST'])
101     def updateBlockPrediction():
102         if not request.json or not 'git_url' in request.json or not '
block_id' in request.json or not 'prediction' in request.json:
103             return jsonify({'message': 'Invalid request'})
104         git_url = request.json['git_url']
105         blockId = request.json['block_id']
106         prediction = request.json['prediction']
107
108         dbms = getDBMS(git_url)
109         dbms.updateBlockPrediction(blockId, prediction)
110
111         return jsonify(
112             {
113                 'message': 'Update success!',
114                 'code': 200,
115                 'success': True,
116             }

```

```

117         )
118
119     @app.route('/updateBlockPrediction', methods=['OPTIONS'])
120     def updateBlockPredictionOptions():
121         print(request.json)
122         print("wrong method")
123         return jsonify({'message': 'Options request'})
124
125     @app.route('/generateTest', methods=['POST'])
126     def generateTest():
127         try:
128             if not request.json or not 'git_url' in request.json or
not 'block_id' in request.json:
129                 return jsonify({'message': 'Invalid request'})
130             git_url = request.json['git_url']
131             blockId = request.json['block_id']
132
133             dbms = getDBMS(git_url)
134             tg = Test_Generator()
135
136             testFileContent = tg.generate_test_case(
137                 package_name= dbms.project.getName(),
138                 code_location=dbms.getBlockOriginalFile(blockId),
139                 function_name_and_arguments=dbms.getBlockName(blockId)
140             ,
141                 prediction=dbms.getBlockPrediction(blockId),
142             )
143             file_name = 'block_' + str(blockId) + '_test.dart'
144
145             dbms.project.create_test(
146                 filename=file_name,
147                 content=testFileContent,
148                 isOverWrite=True
149             )
150             # validation process
151             run_result, run_error = dbms.project.run_test(file_name)
152             iteration_limit = 5
153             while run_error != '' and iteration_limit > 0:
154                 new_test_content = tg.fix_generated_code(
155                     error_message=run_error,
156                     current_test_code=testFileContent,
157                     prediction=dbms.getBlockPrediction(blockId),
158                 )
159                 dbms.project.create_test(
160                     filename=file_name,
161                     content=new_test_content,
162                     isOverWrite=True
163                 )
164                 run_result, run_error = dbms.project.run_test(
165                     file_name)
166                 iteration_limit -= 1
167
168             return jsonify(
169                 {
170                     'message': 'Test generation success!',
171                     'code': 200,
172                     'success': True,
173                     'test_file_content': testFileContent,
174                 }

```

```

173         )
174     except Exception as e:
175         return jsonify({'message': str(e)})
176
177 @app.route('/generateTest', methods=['OPTIONS'])
178 def generateTestOptions():
179     print(request.json)
180     print("wrong method")
181     return jsonify({'message': 'Options request'})
182
183 if __name__ == '__main__':
184     app.run(debug=True)

```

Listing A.15: main.py file.

```

1 class Table:
2     def __init__(self, name: str, columns: dict):
3         self.name = name
4         self.columns = columns
5
6     def getCreateSQL(self):
7         sql = f'CREATE TABLE IF NOT EXISTS {self.name} ('
8         for column in self.columns:
9             sql += f'{column} {self.columns[column]}, '
10        sql = sql[:-2] + ')'
11        return sql
12
13    def getSelectSQL(self, fields: list, conditions: dict):
14        # if conditions is empty, return all
15        res = f'SELECT '
16        if len(fields) == 0:
17            res += '*'
18        else:
19            for field in fields:
20                res += f'{field}, '
21            res = res[:-2]
22        res += f' FROM {self.name}'
23
24        if len(conditions) > 0:
25            res += ' WHERE '
26            for condition in conditions:
27                res += f"{condition} = '{conditions[condition]}' AND "
28            res = res[:-4]
29
30        return res
31    pass
32
33    def getInsertSQL(self, values: dict):
34        sql = f'INSERT INTO {self.name} ('
35        for column in values:
36            sql += f'{column}, '
37        sql = sql[:-2] + ") VALUES ("
38        for column in values:
39            sql += f"'{values[column]}' , "
40        sql = sql[:-2] + ') '
41        return sql
42
43    def getUpdateSQL(self, values: dict, conditions: dict):
44        sql = f"UPDATE {self.name} SET "
45        for column in values:

```

```

46         sql += f"{column} = \'{values[column]}\', "
47     sql = sql[:-2] + " WHERE "
48     for column in conditions:
49         sql += f"{column} = \'{conditions[column]}\', AND "
50     sql = sql[:-4]
51     return sql

```

Listing A.16: Table class.

```

1  @staticmethod
2  def getTable():
3      from DBMS.Table import Table
4      return Table(
5          'BlockType',
6          {
7              'id': 'INT AUTO_INCREMENT PRIMARY KEY',
8              'name': 'VARCHAR(255)'
9          }
10     )

```

Listing A.17: getTable function - BlockType class.

```

1  @staticmethod
2  def getTable():
3      from DBMS.Table import Table
4      return Table(
5          'Block',
6          {
7              'id': 'INT AUTO_INCREMENT PRIMARY KEY',
8              'name': 'VARCHAR(255)',
9              'content': 'TEXT',
10             'prediction': 'TEXT',
11             'type': 'INT',
12             '': 'FOREIGN KEY (type) REFERENCES BlockType(id)'
13         }
14     )

```

Listing A.18: getTable function - Block class.

```

1  @staticmethod
2  def getTable():
3      from DBMS.Table import Table
4      return Table(
5          'ConnectionType',
6          {
7              'id': 'INT AUTO_INCREMENT PRIMARY KEY',
8              'name': 'VARCHAR(255)'
9          }
10     )

```

Listing A.19: getTable function - ConnectionType class.

```

1  @staticmethod
2  def getTable():
3      from DBMS.Table import Table
4      return Table(
5          'Connection',
6          {
7              'id': 'INT AUTO_INCREMENT PRIMARY KEY',

```



```

8         'head': 'INT',
9         'tail': 'INT',
10        'type': 'INT',
11        '': 'FOREIGN KEY (head) REFERENCES Block(id)',
12        '': 'FOREIGN KEY (tail) REFERENCES Block(id)',
13        '': 'FOREIGN KEY (type) REFERENCES ConnectionType(id)',
14    }
15    )

```

Listing A.20: getTable function - Connection class.

```

1  CLASS DBMS:
2      CONSTRUCTOR(project):
3          self.project = project
4
5          # Initialize database if needed
6          IF NOT self._isDBinit() THEN
7              self._initDB()
8          END IF
9
10         # Insert project if not exists
11         IF NOT self._isProjectExistInDB() THEN
12             self._insertProject()
13         END IF
14
15     METHOD getJsonDiagram():
16         # Retrieve diagram data from database
17         Query blocks and connections from DB
18
19         # Format data as JSON structure
20         Create project, blocks, and connections structure
21         Convert block and connection types to string names
22
23         RETURN formatted diagram
24
25     METHOD getBlockName(blockId):
26         Query block name from database
27         RETURN result
28
29     METHOD getBlockContent(blockId):
30         Query block content from database
31         RETURN result
32
33     METHOD getBlockPrediction(blockId):
34         Query block prediction from database
35         RETURN result
36
37     METHOD getBlockOriginalFile(blockId):
38         # Backtrack through connections to find containing file
39         WHILE current block is not a File:
40             Query for parent block
41             Update current block
42         END WHILE
43
44         Extract file path relative to project
45         RETURN path
46
47     METHOD updateBlockPrediction(blockId, prediction):
48         Escape special characters in prediction
49         Update prediction in database

```

```

50
51 METHOD _initDB():
52     # Create required tables
53     Create Project table
54     Create Block table
55     Create Connection table
56     Insert enum values
57
58 METHOD _insertProject():
59     # Insert project record
60     Insert project into database
61
62     # Create and store dependency diagram
63     Create DependencyDiagram
64     Convert blocks and connections to database records
65
66 METHOD _mapBlocksIntoDB(blocks):
67     FOR EACH block IN blocks:
68         Insert block properties into database
69     END FOR
70
71 METHOD _mapConnectionsIntoDB(connections):
72     FOR EACH connection IN connections:
73         Insert connection with block IDs into database
74     END FOR
75 END CLASS

```

Listing A.21: DBMS class (Pseudocode).